

Variance-based adaptive sequential sampling for Polynomial Chaos Expansion

Lukáš Novák^{a,*}, Miroslav Vořechovský^a, Václav Sadílek^a, Michael D. Shields^b

^a Brno University of Technology, Brno, Czech Republic

^b Johns Hopkins University, Baltimore, USA

Received 27 March 2021; received in revised form 5 August 2021; accepted 8 August 2021

Available online 31 August 2021

Abstract

This paper presents a novel adaptive sequential sampling method for building Polynomial Chaos Expansion surrogate models. The technique enables one-by-one extension of an experimental design while trying to obtain an optimal sample at each stage of the adaptive sequential surrogate model construction process. The proposed sequential sampling strategy selects from a pool of candidate points by trying to cover the design domain proportionally to their local variance contribution. The proposed criterion for the sample selection balances both *exploitation* of the surrogate model and *exploration* of the design domain. The adaptive sequential sampling technique can be used in tandem with any user-defined sampling method, and here was coupled with commonly used Latin Hypercube Sampling and advanced Coherence D-optimal sampling in order to present its general performance. The obtained numerical results confirm its superiority over standard non-sequential approaches in terms of surrogate model accuracy and estimation of the output variance.

© 2021 Elsevier B.V. All rights reserved.

Keywords: Polynomial Chaos Expansion; Adaptive sampling; Sequential sampling; Coherence optimal sampling

1. Introduction

The Polynomial Chaos Expansion (PCE), originally proposed by Norbert Wiener [1] and further investigated in the context of engineering problems by many researchers, e.g. [2,3], represents a spectral expansion of the original *stochastic problem* in a polynomial basis. PCE approximation represents very efficient method for sensitivity analysis, uncertainty quantification or reliability analysis [4]. Moreover, once the PCE is available, it is possible to investigate the constructed explicit function in order to estimate additional information about the original problem including its statistical moments, output probability distribution or sensitivity indices without additional sampling [5], which is especially beneficial in industrial applications [6,7]. PCE can be generally formulated in intrusive or non-intrusive form. Despite the recent progress in research on the intrusive approach [8], it is still rarely employed in practical applications since it requires redesign of the mathematical model solver.

On the other hand, the non-intrusive approach offers a convenient way to perform probabilistic analysis of any black-box model. There are generally two types of non-intrusive methods for calculation of deterministic

* Corresponding author.

E-mail addresses: novak.l@fce.vutbr.cz (L. Novák), vorechovsky.m@vut.cz (M. Vořechovský), sadilek.v@fce.vutbr.cz (V. Sadílek), michael.shields@jhu.edu (M.D. Shields).

coefficients: spectral projection and linear regression. The spectral projection approach utilizes the orthogonality of multivariate polynomials and calculates the coefficients using inner products. Although the integrals in spectral projection can be calculated by traditional tensor-product quadrature rules, the number of collocation points grows exponentially with the number of input random variables which is called *curse of dimensionality* and thus computationally far more efficient sparse grids [9,10] should be employed. The second type of the non-intrusive approach is based on linear regression. Although it is typically less expensive than spectral projection (the number of samples should be at least $\mathcal{O}(P \ln(P))$, where P is the number of terms in PCE [11,12]) it suffers from the curse of dimensionality as well, since number of PCE terms is extremely large for high dimensions and high polynomial orders. Therefore, it is necessary to employ advanced adaptive techniques for construction of sparse PCE in order to obtain efficient solutions for real-life physical systems. Moreover, the regression based PCE can be significantly affected by sampling schemes as was recently shown in extensive review paper [13]. Therefore, this paper is focused on the combination of PCE adaptivity with a sequential sampling strategy designed for the non-intrusive approach based on linear regression.

Since each evaluation of a computer model representing the engineering problem is typically highly time-consuming (e.g. non-linear finite element method), it is necessary to reduce the number of model evaluations as much as possible in the process of training the surrogate model, while maintaining the accuracy of the approximation. The balance between accuracy and computational requirements is strongly connected to the selection of the support points in the *design domain* of input variables — a computational design of experiments (DoE). Besides the commonly used crude Monte Carlo sampling, there are several advanced techniques developed in the fields of statistical/numerical estimation of integrals which improve the efficiency and accuracy of the DoE. One of the most widely used techniques is Latin Hypercube Sampling (LHS) [14], a variance reduction technique that uses stratified selection of sampling points. Another popular strategy for DoE is to uniformly fill the design domain according to some *space-filling* criteria such as miniMax, Maximin [15] or generalized versions of distance-based criteria [16,17], or to decrease the *discrepancy* of the point set. Low discrepancy designs can be obtained either by direct algorithmic minimization of selected discrepancy measure or as Quasi Monte Carlo sequences (known also as low-discrepancy sequences, or number-theoretical designs; see e.g. the sequence due to Halton [18,19], Sobol' [20,21], Niederreiter [22–24], Faure [25], the generalization of the Faure sequences by Tezuka [26], and others). These techniques for DoE can be used for general probabilistic analysis (=numerical integration) without any knowledge about the specific mathematical model or surrogate model.

Further it is often beneficial to include additional information into the DoE stemming from the specific type of the surrogate model at hand. The coherence-based sampling was proposed specifically for PCE constructed by ordinary least squares regression (OLS) [27] and it leads to higher stability in estimation of PCE coefficients in comparison to general sampling methods such as LHS, which is commonly used in combination with PCE. Another method developed specifically for OLS is induced sampling [28], which has been proved to be optimal for weighted least-squares methods.

Methods for DoE construction usually need to specify the number of simulations *a priori*. It is however much more efficient and practical to sample additional points one-by-one until desired accuracy of the approximation is reached. Such methodology for sample size extension is referred to as sequential sampling and it is especially beneficial in practical engineering applications. Sequential sampling schemes are often driven by a defined criterion to compare candidates for sample size extension. The concept of adaptive experimental design for learning surrogate models is often termed *active learning*. This approach is a common approach when the goal is reliability analysis with a surrogate: an initial experimental design is iteratively updated based on the current estimation of the limit-state surface in an active learning algorithm [29–31]. Active learning approach involving PCE in the context of reliability analysis was used e.g. in [32–34].

Although there are recent studies focused on general sequential sampling based on space-filling criteria or alphabetical optimality used for PCE [35,36], it is beneficial to use both *exploitation* (leveraging model behavior) criteria and *exploration* (space filling) criteria for definition of an optimally balanced criterion [37]. Such sequential sampling for sparse Bayesian learning PCE combining both aspects — epistemic uncertainty of the statistical inference (exploration) together with quadratic loss function (local exploitation) was recently proposed [38]. However, its application is limited to PCE build by sparse Bayesian learning only. This paper presents a novel adaptive sequential sampling technique with such a balanced criterion. The technique presented in this paper can be coupled with any sparse regression solver and common methods for DoE such as LHS and thus can be easily

implemented into the existing software solutions for PCE construction (e.g. [39–43]). Additionally, in order to increase the efficiency of the proposed scheme, the developed technique is coupled with coherence D-optimal sampling created specifically for non-intrusive PCE solved by OLS and thus all parts of the method are designed in order to increase the efficiency and accuracy of this particular type of surrogate model.

2. Polynomial Chaos expansion

Assume a probability space $(\Omega, \mathcal{F}, \mathcal{P})$, where Ω is an event space, $\mathcal{F}$ is a σ -algebra on Ω (collection of subsets closed under complementation and countable unions) and $\mathcal{P}$ is a probability measure on $\mathcal{F}$. If the input variable of a mathematical model, $Y = g(X)$, is a random variable $X(\omega)$, $\omega \in \Omega$, the model response $Y(\omega)$ is also a random variable. Assuming that Y has a finite variance, PCE represents the output variable Y as a function of an another random variable ξ called the *germ* with given distribution

$$Y = g(X) = g^{\text{PCE}}(\xi), \quad (1)$$

and representing the function $g(X)$ via polynomial expansion in a manner similar to the Fourier series of a periodic signal. A set of polynomials, orthogonal with respect to the distribution of the germ, is used as a basis of the Hilbert space $L^2(\Omega, \mathcal{F}, \mathcal{P})$ of all real-valued random variables of finite variance, where $\mathcal{P}$ takes over the meaning of the probability distribution. The orthogonality condition for all $j \neq k$ is given by the inner product of $L^2(\Omega, \mathcal{F}, \mathcal{P})$ defined for any two functions ψ_j and ψ_k with respect to the weight function p_ξ (probability density function of ξ) as:

$$\langle \psi_j, \psi_k \rangle = \int \psi_j(\xi) \psi_k(\xi) p_\xi(\xi) d\xi = 0. \quad (2)$$

This means that there are specific orthogonal polynomials associated with the corresponding distribution of the germ via its weighting function. For example, Hermite polynomials orthogonal to the Gaussian measure are associated with normally distributed germs. Orthogonal polynomials corresponding to other distributions can be chosen according to Wiener–Askey scheme [44]. For further processing, it is beneficial to use normalized polynomials (orthonormal), where the inner product is equal to the Kronecker delta δ_{jk} , i.e. $\delta_{jk} = 1$ if and only if $j = k$, and $\delta_{jk} = 0$ otherwise

$$\langle \psi_j, \psi_k \rangle = \delta_{jk}. \quad (3)$$

In the case of X and ξ being vectors containing M independent random variables, the polynomial $\Psi(\xi)$ is multivariate and it is built up as a tensor product of univariate orthogonal polynomials as

$$\Psi_\alpha(\xi) = \prod_{i=1}^M \psi_{\alpha_i}(\xi_i), \quad (4)$$

where $\alpha \in \mathbb{N}^M$ is a set of integers called the *multi-index*. The quantity of interest (QoI), i.e. the response of the mathematical model $Y = g(X)$, can then be represented, according to Ghanem and Spanos [3], as

$$Y = g(X) = \sum_{\alpha \in \mathbb{N}^M} \beta_\alpha \Psi_\alpha(\xi), \quad (5)$$

where β_α are deterministic coefficients and Ψ_α are multivariate orthogonal polynomials.

2.1. Non-intrusive computation of PCE coefficients

For practical computation, PCE expressed in Eq. (5) must be truncated to a finite number of terms P . The truncation is commonly achieved by retaining only terms whose total degree $|\alpha|$ is less than or equal to a given p . Therefore, the truncated set of PCE terms is then defined as

$$\mathcal{A}^{M,p} = \left\{ \alpha \in \mathbb{N}^M : |\alpha| = \sum_{i=1}^M \alpha_i \leq p \right\}. \quad (6)$$

The cardinality of the truncated *index set* $\mathcal{A}^{M,p}$ is given by

$$\text{card } \mathcal{A}^{M,p} = \frac{(M+p)!}{M! p!}. \quad (7)$$

Moreover, in practical applications, it is beneficial to prefer only basis functions with lower number of interaction terms. Therefore, it was proposed by Blatman and Sudret [2] to create a PCE basis by a “hyperbolic” truncation scheme:

$$\mathcal{A}^{M,p,q} = \left\{ \alpha \in \mathbb{N}^M : \|\alpha\|_q \equiv \left(\sum_{i=1}^M \alpha_i^q \right)^{1/q} \leq p \right\}. \quad (8)$$

Note that selection of $q = 1$ corresponds to the standard truncation scheme according to Eq. (6) and, for $q < 1$, terms representing higher-order interactions are eliminated. Such an approach leads to a dramatic reduction in the cardinality of the truncated set for high total polynomial orders p and high dimensions M .

When PCE is truncated to a finite number of terms, there is an error ε of the approximation such that

$$Y = g(X) = \sum_{\alpha \in \mathcal{A}} \beta_{\alpha} \Psi_{\alpha}(\xi) + \varepsilon.$$

From a statistical point of view, PCE is a simple linear regression model with intercept. Therefore, it is possible to use *ordinary least square* (OLS) regression to minimize the error ε

$$\beta = \arg \min_{\beta \in \mathbb{R}^P} \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} [\beta^T \Psi(\xi^{(i)}) - g(x^{(i)})]^2. \quad (9)$$

Knowledge of vector β fully characterizes the approximation via PCE. To solve for β , first it is necessary to create n_{sim} realizations of the input random vector X and the corresponding results of the original mathematical model $\mathcal{Y}$, together called the experimental design (ED). Then, the vector of P deterministic coefficients β is calculated as

$$\beta = (\Psi^T \Psi)^{-1} \Psi^T \mathcal{Y}, \quad (10)$$

where Ψ is the data matrix

$$\Psi = \{ \Psi_{ij} = \Psi_j(\xi^{(i)}), i = 1, \dots, n_{\text{sim}}, j = 0, \dots, P-1 \}. \quad (11)$$

Note that the number of terms P is highly dependent on the number of input random variables M and the maximum total degree of polynomials p . Estimation of β by regression then needs at least the number of samples $\mathcal{O}(P \ln(P))$ for stable solution [11,12]. Therefore, in case of a large stochastic model, the problem can become computationally highly demanding. However, one can utilize advanced model selection algorithms such as Least Angle Regression (LAR) [45] to find an optimal set of PCE terms and thus reduce the number of samples needed to compute the unknown coefficients if the true coefficient vector is sparse or compressible as proposed by Blatman and Sudret [2]. Note that beside LAR, there are other best model selection algorithms such as orthogonal matching pursuit [46] or Bayesian compressive sensing [47] with comparable numerical results. The sparse set of basis functions obtained by any adaptive algorithm is further denoted for the sake of clarity as $\mathcal{A}$.

2.2. Estimation of approximation error

Once the PCE is constructed, it is crucial to estimate its accuracy. Further, the accuracy of PCE can be used for the direct comparison among several PCEs in order to choose the best surrogate model. Therefore it is beneficial to use methods which do not need any additional sampling of the original mathematical model. A common choice is the coefficient of determination R^2 , which is well known from machine learning. However, R^2 may lead to overfitting and thus advanced methods should be used. One of the most utilized methods for measuring the performance of the learning algorithm in recent years is the leave-one-out cross validation error Q^2 . This statistic is based on residuals between the original surrogate model and the surrogate model built with the ED while excluding one realization. This approach is repeated for all realizations in the ED and the average error is estimated. Although the calculation of Q^2 is typically highly time-consuming, it is possible to obtain results analytically from a single PCE as follows [48]

$$Q^2 = 1 - \frac{\frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \left[\frac{g(x^{(i)}) - g^{\text{PCE}}(x^{(i)})}{1 - h_i} \right]^2}{\sigma_{Y,\text{ED}}^2}, \quad (12)$$

where $\sigma_{Y,ED}^2$ is a variance of experimental design calculated using the original mathematical model and h_i represents the i th diagonal term of matrix $\mathbf{H} = \Psi (\Psi^T \Psi)^{-1} \Psi^T$.

2.3. Statistical moments derived from PCE

The specific form of PCE together with the orthogonality of the polynomials allows for a powerful and efficient post-processing. Once a PCE approximation is created, it is possible to obtain statistical moments of the QoI. Generally, its raw statistical moment of the m th order is defined as

$$\begin{aligned} \langle Y^m \rangle &= \int [g(X)]^m p_X(X) dX = \int \left[\sum_{\alpha \in \mathbb{N}^M} \beta_\alpha \Psi_\alpha(\xi) \right]^m p_\xi(\xi) d\xi \\ &= \int \sum_{\alpha_1 \in \mathbb{N}^M} \dots \sum_{\alpha_m \in \mathbb{N}^M} \beta_{\alpha_1} \dots \beta_{\alpha_m} \Psi_{\alpha_1}(\xi) \dots \Psi_{\alpha_m}(\xi) p_\xi(\xi) d\xi \\ &= \sum_{\alpha_1 \in \mathbb{N}^M} \dots \sum_{\alpha_m \in \mathbb{N}^M} \beta_{\alpha_1} \dots \beta_{\alpha_m} \int \Psi_{\alpha_1}(\xi) \dots \Psi_{\alpha_m}(\xi) p_\xi(\xi) d\xi. \end{aligned} \quad (13)$$

As can be seen from the final part of the formula, in case of PCE, it is necessary to integrate only over the basis functions (orthonormal polynomials), which leads to a dramatic simplification in comparison to the integration of the original mathematical function. Moreover, it is also possible to write an analytical expression of the integral in several cases. Specifically, the first statistical moment (mean value) is obtained as

$$\mu_Y = \langle Y^1 \rangle = \int \left[\sum_{\alpha \in \mathbb{N}^M} \beta_\alpha \Psi_\alpha(\xi) \right] p_\xi(\xi) d\xi = \sum_{\alpha \in \mathbb{N}^M} \beta_\alpha \int \Psi_\alpha(\xi) p_\xi(\xi) d\xi. \quad (14)$$

Considering the orthonormality of the polynomials

$$\int \Psi_\alpha(\xi) p_\xi(\xi) d\xi = 0 \quad \forall \alpha \neq 0, \quad \Psi_0 \equiv 1,$$

it is possible to obtain the mean value directly from the PCE deterministic coefficients. Namely, the mean value is equal to the first deterministic coefficient of the expansion

$$\mu_Y = \langle Y^1 \rangle = \beta_0. \quad (15)$$

The second raw statistical moment, $\langle Y^2 \rangle$, is written as

$$\begin{aligned} \langle Y^2 \rangle &= \int \left[\sum_{\alpha \in \mathcal{A}} \beta_\alpha \Psi_\alpha(\xi) \right]^2 p_\xi(\xi) d\xi = \sum_{\alpha_1 \in \mathcal{A}} \sum_{\alpha_2 \in \mathcal{A}} \beta_{\alpha_1} \beta_{\alpha_2} \int \Psi_{\alpha_1}(\xi) \Psi_{\alpha_2}(\xi) p_\xi(\xi) d\xi \\ &= \sum_{\alpha \in \mathcal{A}} \beta_\alpha^2 \int \Psi_\alpha(\xi)^2 p_\xi(\xi) d\xi = \sum_{\alpha \in \mathcal{A}} \beta_\alpha^2 \langle \Psi_\alpha, \Psi_\alpha \rangle. \end{aligned} \quad (16)$$

Considering again the orthonormality of the polynomials, defined by the inner product in Eq. (3), it is possible to obtain the variance $\sigma_Y^2 = \langle Y^2 \rangle - \mu_Y^2$ as the sum of all squared deterministic coefficients except the intercept (which represents the mean value), i.e.

$$\sigma_Y^2 = \sum_{\substack{\alpha \in \mathcal{A} \\ \alpha \neq 0}} \beta_\alpha^2. \quad (17)$$

Note that, higher statistical central moments skewness γ_Y (3rd moment) and kurtosis κ_Y (4th moment) need precomputing of the triple and quadruple products.

In the following, we select the variance of the output variable, i.e. σ_Y^2 in Eq. (17), as the *target characteristic* of Y and we focus on development of the sequential sampling strategy in order to estimate this variance as accurately as possible at any stage of PCE build. The reason for selection of variance is that we expect a monotonic relationship between function *variation* in the sense of Hardy and Krause [49] and its *variance*. As will be shown below, Eq. (16) can be modified in order to obtain the local contribution to variance. We conjecture that it is therefore

important to place samples densely in regions of large local variance and sparsely in regions of small local variance, in order to obtain a near-optimal sample [37]. This arrangement of samples may decrease the error of the function approximation and its integral.

3. Sampling methods

Assuming a non-intrusive approach for calculation of the PCE coefficients using OLS defined in Eq. (9), it is necessary to create an ED containing n_{sim} realizations of the input random vector and the vector of corresponding results of the original mathematical model. Typically we consider sampling of ξ using its density f_{ξ} , which represents the distribution according to the Wiener–Askey scheme. For the M -dimensional Legendre polynomials this means sampling uniformly from the M -dimensional hypercube $[-1, 1]^M$ and for M -dimensional Hermite polynomials, it corresponds to sampling from the M -dimensional Gaussian distribution with independent standardized Gaussian marginal distributions. Naturally, it is crucial to use an efficient sampling scheme for the DoE of ξ in order to obtain accurate results for a given computational budget.

DoE has been an area of interest for many researchers since the beginning of uncertainty quantification and structural reliability. The most simple but generally applicable method is crude Monte Carlo Sampling (MC), i.e. a method associated especially with robust (and inefficient) numerical integration. In standard Monte Carlo integration, the important condition of integration being unbiased is that the sample is selected *uniformly* and *independently* with respect to the target density. In MC the sampling points are selected independently of each other and therefore, clusters of points emerge randomly as well as regions which are not covered by any point.

There has been considerable effort spent on improving the spatial arrangement of points in a sample. The Koksma–Hlawka inequality [49], which was developed to predict an upper bound of integration error, motivates the decrease in *discrepancy* of the sample set (ED). Discrepancy in a way measures *uniformity* of a point set, i.e. the difference between the desired uniform distribution and the empirical distribution of the point set. Such uniformity of point distribution may be useful also for initial screening or building a surrogate model. Low discrepancy can be achieved either by direct minimization of a suitable discrepancy measure [17,50–52] or simply by using various Quasi Monte Carlo sequences mentioned in the Introduction section. Quasi Monte Carlo sequences are deterministic point sets and they allow for sequential addition of points one-by-one while retaining an optimal rate of star discrepancy decrease with increasing sample size [53].

Another branch of research focuses on variance reduction techniques such as *importance sampling* which place points according to a predefined or adaptively adjusted *sampling* density which can be different from the target density or methods of stratified selection of sampling points that improve spatial arrangement of the sampling points. One of the most widely used techniques is Latin Hypercube Sampling (LHS) first suggested by Conover [14]; see also [54]. LHS specifically has the effect of reducing variance associated with the additive components of a transformation. Hence, for functions that are dominated by the main effects of the individual variables, LHS will significantly reduce variance. For functions with significant variable interactions, it is less effective. [55–58].

Another important aspect of sample selection for DOE is the uniformity of filling of the design domain. This aspect is important for *function approximation* and resulted in the seminal works on *space-filling* criteria based on mutual distances among points: the Maximin and miniMax criteria [15]. These criteria prefer designs without point clusters or without large empty areas. A generalization of the Maximin criterion to the phi-criterion [16] was presented along with a heuristic construction algorithm that can be combined with the LHS strategy. Similarly, the miniMax criterion can be generalized [59]. Moreover, the recently developed *periodic* versions of the whole class of distance-based criteria [17,60] guarantee *statistically uniform* distribution of points along with *even point distribution* within each single design. These distance-based criteria can be employed for direct design optimization.

Uniformity of ED and space-filling criteria are important characteristics to obtain a quality ED for PCE. The LHS thus typically leads to more accurate estimation of β in comparison to crude MC [61,62]. Sampling for PCE, however, might be motivated by additional criteria of optimality than the space-filling property, discrepancy or statistical uniformity with respect to the target distribution. The optimality criterion should consider aspects related to the particular method of identification of the polynomial modes and their coefficients.

This naturally leads to sampling from a probability measure different from the input distributions, which minimize a selected characteristic of the ED. Moreover, it is also necessary to modify the basis functions in order to preserve orthonormality of the data matrix. Minimizing the coherence parameter of a PCE basis functions [63] leads to the coherence optimal and related asymptotic sampling [27], which is theoretically more efficient in comparison

to standard sampling based on the input distribution. Similarly the Christoffel sparse approximation [64] was derived using a different definition of the coherence parameter. Note that these sampling strategies are derived for a specific purpose (non-intrusive PCE) and thus their efficiency in general probabilistic analysis might be unsatisfying. Although coherence optimal sampling is more computationally demanding, it is not a crucial problem in engineering, where the calculation of mathematical models takes the largest part of the whole process.

Also, it would be optimal if the PCE that has been set at any stage of sampling based on the available information about the samples and the corresponding function values $g(\mathbf{x})$, can propose the new sampling point (sample size extension). With such an algorithm at hand, one can build the PCE approximation incrementally while exploiting all the knowledge available so far. For efficient accurate exploitation, it is necessary to use an adaptive algorithm for PCE construction (by *adaptive* we mean selection of the most important combinations of modes in the index set $\mathcal{A}$). There are several methods for adaptive selection of the optimal PCE basis functions such as LAR [45] employed in numerical experiments. The selected basis functions are further used for the process of exploitation and thus it is crucial to identify new basis functions in every iteration of the sequential sampling in order to obtain the best possible location for a new sampling point. Moreover, the adaptivity of the basis functions is important for the accurate coherence optimal sampling briefly described in the following paragraphs.

3.1. Coherence-optimal sampling

Generally, it is beneficial to take all pieces of information about the given mathematical task into account in order to choose a correct methodology. Although standard sampling based on the input distribution is suitable for any probabilistic analysis, there are more efficient methods developed specifically for PCE solved by over-determined OLS [27], which is employed in this paper.

Coherence-optimal sampling constructs a new sampling measure minimizing a coherence parameter associated with stability and convergence of the PCE solved by OLS. The coherence parameter $\mu(Y)$ is defined as [65]

$$\mu(Y) := \sup_{\xi} \sum_{j=1}^P |w(\xi)\psi_j(\xi)|^2, \quad (18)$$

where the weight function $w(\xi)$ is

$$w(\xi) := \frac{P}{B(\xi)}. \quad (19)$$

The analytical expression of $B(\xi)$ is generally not available, but it is possible to evaluate its value for arbitrary ξ as

$$B(\xi) = \sqrt{\sum_{j=1}^P |\psi_j(\xi)|^2}. \quad (20)$$

Finally, the coherence-optimal probability measure is defined as

$$f_{\text{coh}}(\xi) := P^{-1} f(\xi) B^2(\xi). \quad (21)$$

In order to sample from $f_{\text{coh}}(\xi)$ one may use Markov Chain Monte Carlo (MCMC) [65]. The proposal distribution for MCMC is suggested for case when $p \leq M$ where one should use standard sampling from distributions naturally orthogonal to the employed polynomial basis as already described for standard sampling. If $p \geq M$, samples should be independently drawn from a uniform distribution on a M -dimensional ball of radius $\sqrt{2\sqrt{2p+1}}$ for Hermite polynomials, and a M -dimensional Chebyshev distribution for Legendre polynomials.

Note that the coherence-optimal sampling does not generally lead to the orthonormal columns of the data matrix and the deviation $\langle \Psi_{\alpha_1}, \Psi_{\alpha_2} \rangle - \delta_{\alpha_1 \alpha_2}$ may be significant. Therefore, one has to apply a weight function $w(\xi)$ to the basis functions and use weighted least squares in order to estimate β using coherence-optimal sampling as described in previous paragraphs. Weighted least squares method is a special case of generalized least squares containing only diagonal members of the correlation matrix of residuals. Optimal values of the deterministic coefficients is then obtained by solution of the following system of equations:

$$\mathbf{W} \Psi \beta \approx \mathbf{W} \mathcal{Y}, \quad (22)$$

where matrix $\mathbf{W}$ is a diagonal matrix with $W_{i,i} = w(\xi^{(i)})$.

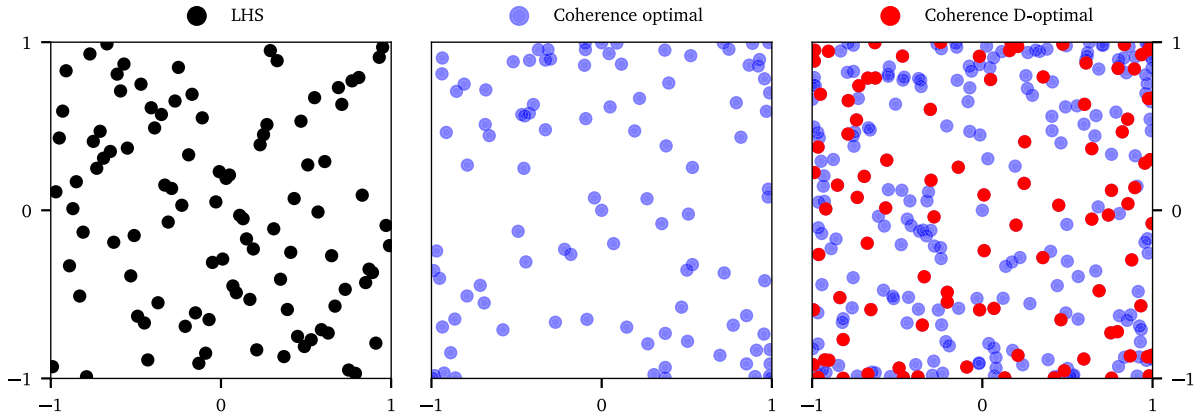


Fig. 1. An ED containing 100 samples generated by LHS, coherence-optimal sampling and coherence D-optimal sampling for Legendre polynomials.

3.2. D-optimal experimental design

Optimality of the ED for OLS can also be measured by the so-called alphabetic criteria of the information matrix $\mathcal{I} := \frac{1}{n_{\text{sim}}} \Psi^T \Psi$, which is crucial for stability of OLS. There are several types of criteria focused on various characteristics of the information matrix; see [62] for a review of criteria in the context of PCE. Although one of the most promising is the S-optimal criterion [35], it is also highly computationally demanding especially for large M . Therefore, in this paper we use the following cheaper and well known estimation-oriented criterion D-optimal design, which is focused on accurate estimation of β .

Since the PCE basis functions are orthonormal, $\mathcal{I}$ is on average identity, but for finite sample size n_{sim} there is a deviation from the identity matrix

$$\|\mathcal{I} - \mathbf{I}\| > 0. \quad (23)$$

The D-optimal design, obtained by maximizing the determinant of the information matrix, leads to small deviation and thus stable estimation of β . For the practical construction of D-optimal ED, it is possible to employ Fedorov exchange algorithm [66], greedy algorithm [67] or rank revealing QR decomposition [68] to find an optimal information matrix containing n_{sim} rows out of a candidate pool containing $n_{\text{pool}} \gg n_{\text{sim}}$ rows.

3.3. Coherence D-optimal experimental design

As originally proposed together with coherence optimal sampling [68], it is beneficial to merge the previous two techniques in order to obtain a *Coherence D-optimal* (Coh D-opt) ED for the stable and accurate solution of WLS. Although Coh D-opt was originally proposed for compressed sensing in the context of PCE [68], D-optimality was already employed in combination with standard sampling (LHS and MC) for OLS and LAR in [35]. Therefore, Coh D-opt can similarly be used also for the non-intrusive approach based on WLS and LAR as employed in this paper. The pool of candidates is generated by coherence optimal sampling and further reduced by D-optimality criterion which leads to uniform ED without clusters of samples as can be seen in Fig. 1.

4. Adaptive sequential sampling

In industrial applications, it is often not feasible to perform a large number of evaluations of the original mathematical model (e.g. FEM) and thus it is important to reduce the number of simulations as much as possible. An efficient approach therefore is *adaptive sequential sampling*, which uses iterative selection of the new sampling points according to specific criteria while exploiting the already available information. Although general sequential sampling is an area of interest for many researchers [69–72], there is still a lack of studies focused specifically on PCE. The recent study [35] compared several simple sequential sampling methods based on D/S-optimality of samples or maximin criterion of samples.

Note that there are two different strategies for sequential sampling. The first is to enrich the initial ED according to a space-filling criterion (exploration) without assuming any knowledge of the mathematical model or PCE form; see e.g. [71,73]. The motivation is clear: we do not want to locate an augmented point very close to an existing point to avoid getting redundant information in the nearby region. However, by obtaining data sequentially, it is possible to learn from the early stages to inform subsequent data collection, minimize wasted resources, and provide answers for various objectives (exploitation). Therefore, the second strategy works with the structure of the PCE (basis functions) in order to identify an optimal sample. Unfortunately, in situations when the initial screening overlooks a globally important region, the exploitation criterion may continue refinement of some other, locally important region that was detected, and there is a risk of never discovering a globally important region. Therefore, it is beneficial to include a balance between both criteria in search for a suitable candidate. Note that, such approach was employed e.g. in [37] in a different context: a criterion motivated by the Koksma–Hlawka inequality [49] was proposed and coupled with stratified sampling in order to improve the efficiency of statistical integration.

As discussed above, the *adaptivity* feature of the PCE surrogate model can be ensured by any model selection algorithm. Moreover, it can be combined also with hyperbolic truncation according to Eq. (8), which is efficient for high P . A general adaptive sequential algorithm thus should adaptively reconstruct the PCE using model selection algorithms in order to identify a sparse set of basis functions $\mathcal{A}$ in each iteration.

The *sequential* feature can be added by using a comparison criterion for selection of the best candidate from a pool of candidates while balancing between exploration and exploitation. *Exploitation* of the local areas of the design domain is focused on identification of sub-domains associated with a defined characteristic of the mathematical model such as high gradient, local maxima etc. The candidates from the identified sub-domains are further preferred. Another typical example can be identification of sub-domains associated with high variation of the mathematical model. Although exploitation is a powerful technique for identification of the best candidate, it is typically based on a built surrogate model and thus it is highly dependent on the quality of a given ED. On the other hand, *exploration* assures uniform coverage of the whole design domain, possibly with respect to specific characteristic as in case of alphabetical optimality [74], and it assures that the algorithm does not get stuck in local minima. It opens the door to detection and exploration of important areas in design domain, where the behavior of the studied function $g(\mathbf{x})$ might be significantly different from possibly incorrect expectation based on the surrogate.

4.1. The proposed Θ criterion for sequential sampling

We propose an adaptive sequential sampling strategy accompanied by a criterion designed for efficient and accurate estimation of β using least squares. Consider a pool of candidates containing n_{pool} realizations of the random vector ξ generated by an arbitrary sampling technique. Once the pool of candidates conditioned by the selected PCE basis is generated, it is necessary to construct a criterion for the selection of the best candidate balancing between the exploitation and exploration of the design domain. Such criterion, called the Θ criterion from here on, is proposed as follows

$$\Theta(\xi^{(c)}) \equiv \Theta_c = \underbrace{\sqrt{\sigma_{\mathcal{A}}^2(\xi^{(c)}) \cdot \sigma_{\mathcal{A}}^2(\xi^{(s)})}}_{\text{ave variance density}} \underbrace{l_{c,s}^M}_{\text{vol.}} \equiv \sqrt{\sigma_c^2 \cdot \sigma_s^2} l_{c,s}^M. \quad (24)$$

where we introduce an abbreviated notation by dropping the point designation $\xi^{(c)}$ and using simply the lower index instead and also by dropping the set of basic functions, $\mathcal{A}$, which is selected for every instant of the algorithm and thus may differ as the sample size increases. The criterion has an intuitive meaning and also has units of *variance* and is a product of two parts: the exploitation part (denoted as “ave variance density”) and the exploration part (the distance term $l_{c,s}$ raised to the domain dimension). Multiplication of these two independent contributions maintains the optimal balance between exploration and exploitation.

The *exploration* aspect is maintained by accounting for the distance $l_{c,s}$ between a candidate $\xi^{(c)}$ and its nearest neighboring point from the existing ED, $\xi^{(s)}$. For the distance term a suitable metric must be selected. In this paper, we select the Euclidean distance between the candidate and its nearest neighbor as

$$l_{c,s} = \sqrt{\sum_{i=1}^M |\xi_i^{(c)} - \xi_i^{(s)}|^2}, \quad (25)$$

although other distances can be considered, particularly in high dimension.

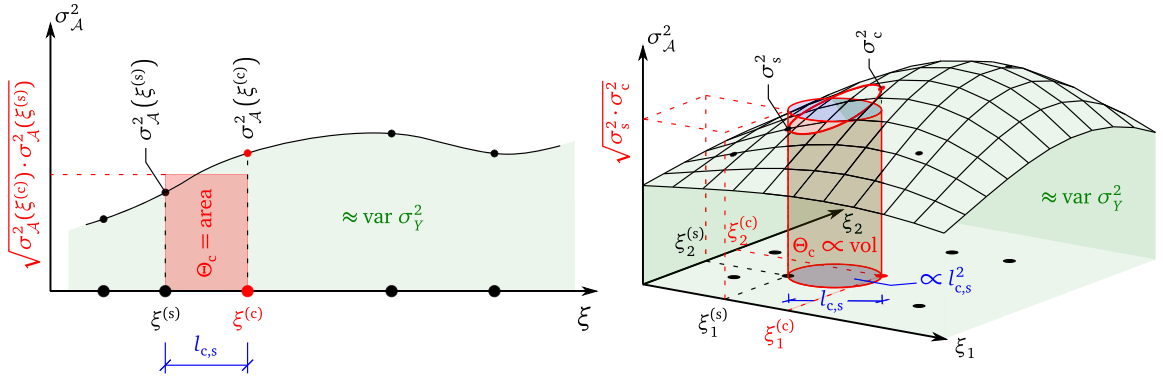


Fig. 2. Geometrical meaning of the proposed Θ criterion for a candidate “c” in two and three dimensions. Black solid circles are existing points and point “s” is the nearest neighbor to candidate “c”. The solid thin curve/surface represents the current estimation of variance density over the design domain.

The *exploitation* in candidate selection is motivated by our desire to *uniformly cover local contributions to the total variance*, σ_Y^2 . By recalling Eq. (17), we know that σ_Y^2 can be estimated simply as the sum of all squared deterministic coefficients except the intercept. The mean square can be obtained as the integral featuring the selected polynomial basis over the design domain; see Eq. (16). This means that the variance can be thought of as an integral of local contributions over the design domain indexed by coordinates ξ . In other words, we need to integrate a local *variance density* $\sigma_A^2(\xi)$. Once the PCE has been established at any given stage of the algorithm, the variance density is computationally cheap to evaluate for any location ξ as

$$\sigma_A^2(\xi) = \left[\sum_{\substack{\alpha \in \mathcal{A} \\ \alpha \neq 0}} \beta_\alpha w(\xi) \Psi_\alpha(\xi) \right]^2 p_\xi(\xi). \quad (26)$$

The local variance is therefore estimated based on the basis functions and coefficients β of the PCE. Depending on technique utilized for sampling of candidates, one should apply the weight $w(\xi)$ (also used in weighted least squares) to basis functions in order to reflect influence of sampling from a probability measure different from the input distribution. Specifically in this paper, $w(\xi)$ is defined for Coh D-opt according to Eq. (19) and $w(\xi) = 1$ for LHS. When considering a candidate “c”, one might think about the variance contribution of the region between the candidate and its nearest neighbor. A rough estimation may be obtained by considering an average of local variance densities between the candidate and its nearest neighbor, “s”. This average is represented by the *geometric mean* between the two numbers. The geometric mean between n numbers x_i is defined as $(\prod_{i=1}^n x_i)^{1/n}$ and therefore, we take the geometric mean of two local variance densities simply as the square root of the local variance densities of the candidate and its nearest neighbor, see the first term in Eq. (24). When this geometric mean is multiplied by the M th power of the distance between the two points, $l_{c,s}^M$, the volume (variance contribution) of a neighborhood between them is estimated; see the sketch in Fig. 2. In other words, the criterion estimates the amount of variance in a “bite” by the candidate. Note that Eq. (24) defines the bite as a hypercube of side-length $l_{c,s}$. However, other geometric entities may be considered without practical impact on the algorithm. The reason is that all the geometric volumes for various candidates under comparison would have the same positive multiplier of $l_{c,s}^M$ which can be dropped as it does not change the ranking of the compared candidates. Therefore, we can say that the proposed criterion helps to select a candidate with roughly the largest amount variance being refined. The balance between exploration and exploitation is maintained: a candidate which is close to an existing point can only be selected if the corresponding variance density is large. Similarly, when a region with low contribution is being detected by the PCE, candidates from such regions are ignored.

In situations when the variance density is a constant function, the criterion collapses to a simple space-filling criterion (a form of miniMax criterion); [15,60]. Such criterion ensures the preference of candidates filling the largest empty regions in the design domain and thus leading to uniform distribution of points in the sense of miniMax design criterion. We remark that miniMax design is a preferable choice for the construction of emulators because it minimizes the worst case prediction variance.

A question may arise: why do we propose to use the geometrical mean instead of arithmetical mean? The reason is that the criterion may also be used for infinite design domains (for example in conjunction with Gaussian germ). In such situation, the pool of candidates may contain points $\xi^{(c)}$ that are very far from the mean value and such a point may have (almost) zero variance density. Yet, the criterion would prioritize it due to the large distance from the nearest neighbor $\xi^{(s)}$ as the arithmetical average with its variance density would equal one half of the two local densities. In other words, infinitely distant candidate points would always win the comparison, despite a vanishing contribution of one of them. Using the geometrical mean prevents unimportant distant points with zero density from being selected.

Maximization of the criterion leads to the best candidate, which is added to active ED. As can be seen, the proposed criterion prefers candidate points in parts of multidimensional space associated with higher contribution to the variance of the mathematical model. This idea is similar to the sequential sampling proposed in [37] based on Koksma–Hlawka inequality respecting both variation of the function and discrepancy of realizations, however the proposed criterion is constructed specifically for PCE and thus it can use the PCE basis functions in order to increase the efficiency of the computation. The significant advantage of the proposed method is the ability to add candidates into existing ED one-by-one and thus it can be employed at any moment of the PCE construction process and it can be combined with any sampling algorithm for construction of initial ED marked with subscript as ξ_{ED} , $\mathbf{W}_{ED}$, $\mathcal{V}_{ED}$.

4.2. Adaptive sequential sampling with LHS-based candidates

LHS represents perhaps the most common sampling technique in surrogate modeling and it can be easily coupled with the proposed sequential sampling in a simple manner. The pool of candidates is generated by LHS and the proposed criterion is employed for the selection of the best candidate. This process is repeated at every iteration of the sequential sampling (with the pool being either regenerated or reused from the preceding step). In order to illustrate the proposed sequential adaptive algorithm, we selected five iterations and depict the corresponding states in Fig. 3. The initial ED is represented by solid black circles and the sequentially selected realizations are plotted using solid red circles. The color maps represent the value of the proposed criterion (right column) and also its individual components (the preceding columns). Since the generation of the pool of candidates by LHS is simple and fast, LHS-based adaptive sequential sampling represents an efficient extension to non-sequential LHS and it could be easily implemented into existing software tools.

4.3. Adaptive sequential sampling with coherence-based candidates

The proposed criterion can generally be coupled with any sampling technique. However, since coherence-based sampling is highly affected by the set $\mathcal{A}$, it might be ideal if the pool of candidates is generated by coherence-based sampling. Further, in order to obtain stable estimates of β , the pool of candidates should be reduced using D-optimality criterion calculated by QR factorization with column pivoting [68] also called rank revealing QR factorization (RRQR). Note that it is necessary to evaluate the proposed criterion for every candidate and thus it might be computationally demanding for large n_{pool} . Therefore, we propose to generate the pool of candidates by coherence D-optimal sampling in every loop of sequential sampling instead of a single large pool generated before the start of the iteration process. Smaller pools for every iteration are not only computationally efficient but such approach reflects the actual sparse set of basis functions of PCE obtained by LAR in each iteration. This might be crucial for the candidate set generated by coherence-based sampling, since it is optimized for the selected basis functions in each step. Moreover, the proposed criterion gives higher importance to basis functions associated to higher β , which is important for identification of functional extremes.

Algorithm 1 thus reflects all pieces of information about the stochastic model (probabilistic distribution of input variables), investigated mathematical model (sparse set of basis functions) and even type of solver for PCE construction (weighted least squares) in order to obtain accurate and stable estimates of the deterministic PCE coefficients β . Combination of all techniques used in the algorithm thus leads to superior performance as will be shown in the numerical examples. However, generation of the pool of candidates is much more computationally demanding in comparison to the LHS-based adaptive sequential sampling.

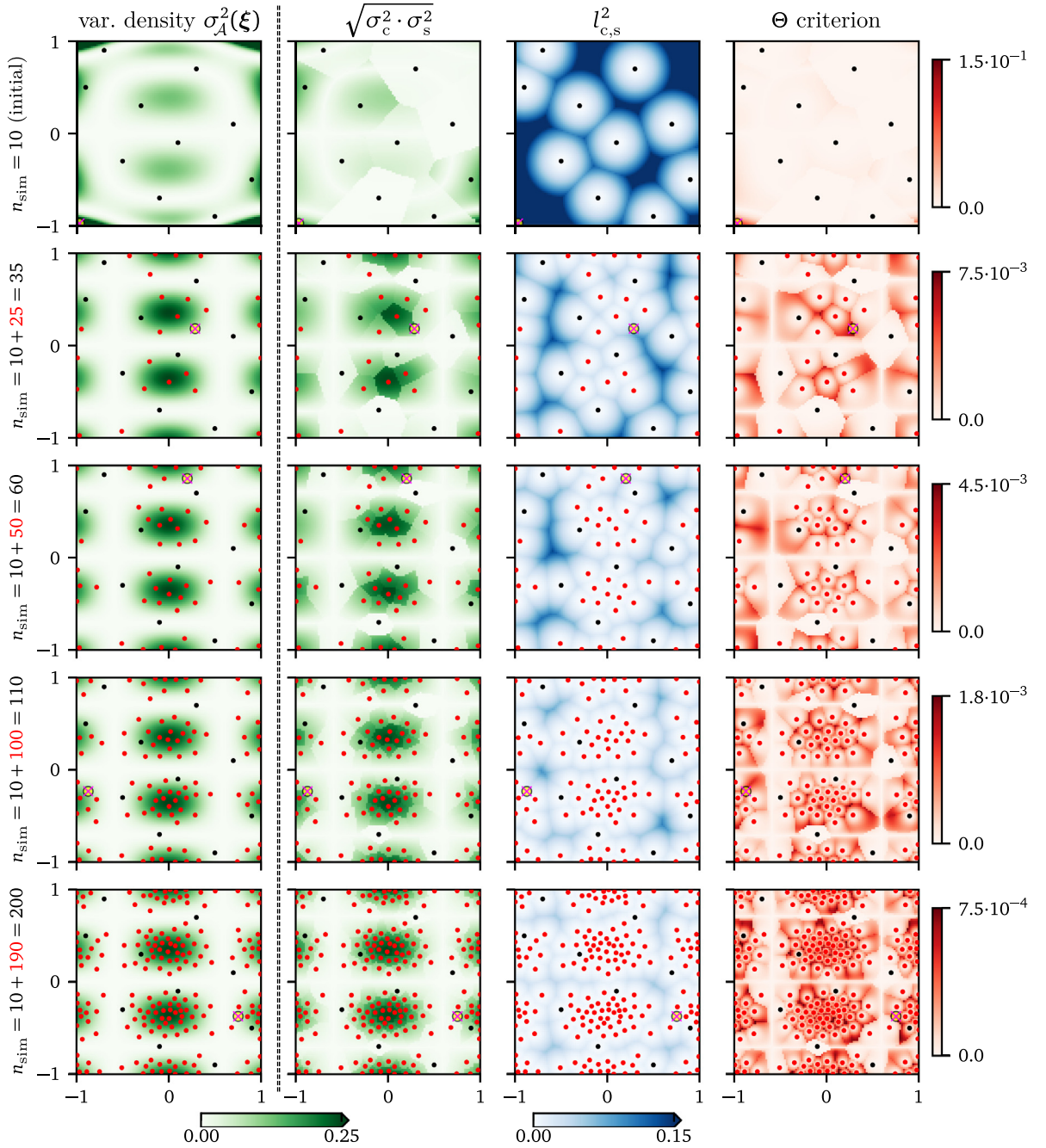


Fig. 3. Illustration of five stages during the proposed sequential sampling. Black solid circles: initial design. Red circles: extended sample. Crossed empty circle: the best candidate. The value of the proposed criterion and its components are depicted using the underlying color maps. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5. Numerical experiments

The proposed algorithm was numerically tested on several examples of increasing complexity. The setup common to all examples was as follows: PCE is solved by non-intrusive OLS (LHS) or WLS (Coh D-opt), a sparse set of the basis functions $\mathcal{A}$ is obtained by LAR with maximum total polynomial order $p = 10$, if not said otherwise.

Algorithm 1 Coherence-Based Adaptive Sequential Sampling — one iteration

Input: $\xi_{ED}, \mathbf{W}_{ED}, \mathcal{Y}_{ED}, \mathcal{A}, \beta, (n_{sim})$ 1: $n = 5P, n_{pool} = 3P$ 2: $\xi_{coh} \leftarrow n$ samples from $f_{coh}(\xi)$ using MCMC (Eq. (21)) 3: $\mathbf{W}_{coh} \leftarrow$ weights corresponding to ξ_{coh} (Eq. (22)) 4: $\xi_{pool}, \mathbf{W}_{pool} \leftarrow n_{pool}$ D-optimal samples & weights 5: for all $\xi^{(c)} \in \xi_{pool}$ do 6: $\xi^{(s)} \leftarrow \arg \min_{\xi \in \xi_{ED}} l_{c,s}(\xi^{(c)}, \xi)$ 7: $\Theta_c = \Theta(\xi^{(c)}, \xi^{(s)})$ 8: end for 9: $\xi^{(new)} \leftarrow \arg \max_{\xi^{(c)}} \Theta_c$ Output: $\xi^{(new)}, w(\xi^{(new)})$	current Experimental Design and the corresponding PCE set the pool sizes based on the p and M draw points from the coherence density calculate the corresponding weights preselect the final pool of candidates loop thru all candidates find the nearest neighbor compute the criterion (Eq. (24)) select the best candidate return the best candidate and the corresponding weight
--	---

Identical p for all examples simulates a possible engineering situation with a black-box function (e.g. finite element analysis) where it is not possible to select the best p a priori. The initial ED for the PCE construction before the first step of the proposed iterative algorithm is generated by LHS and it contains an initial screening design with $n_{sim} = 10$ realizations of the input random vector for the first three examples and $n_{sim} = 20$ for the last example.

Although the proposed criterion can be coupled with any sampling technique for ED generation, only two selected techniques (LHS and Coh D-opt) were employed and compared in the numerical examples. LHS was selected for this study as it is the most common sampling technique for surrogate modeling due to its efficiency and simplicity. Existing software applications and packages for PCE construction (e.g. [39–43]) usually contain implementation of LHS and thus the process can be easily extended by proposed selection criterion. On the other hand, Coh D-opt is not a common approach, thus it is representative of advanced sampling methods suited specifically for least-squares PCE. Coh D-opt EDs usually achieve higher accuracy but their implementation is not straightforward.

Each example is solved by three types of strategies:

- non-sequential approach (non-seq) with ED generated via LHS for each sample size at once — this represents the most common approach employed in surrogate modeling,
- the proposed sequential sampling with candidates generated by LHS, and
- the proposed sequential sampling with candidates generated by Coh D-opt.

The sample size in the first (LHS) strategy is fixed and we study the PCE behavior for a range of these sample sizes selected a priori.

The two sequential sampling strategies differ in the way the pool of candidates is proposed. For the sake of clarity, the pool of candidates obtained by Coh D-opt for sequential sampling contains $3P$ D-optimal samples, which are selected from a greater pool of $5P$ Coh-optimal samples; for LHS sequential sampling, it contains $3P$ samples generated by LHS.

The results are compared in terms of the (i) *relative error in variance* of QoI

$$\epsilon = \frac{|\sigma^2 - \sigma_Y^2|}{\sigma_Y^2}, \quad (27)$$

defined as the absolute deviation of the estimated variance σ^2 from the exact value σ_Y^2 divided by the exact variance, and the (ii) *leave-one-out error* of PCE approximation Q^2 according to Eq. (12). In order to get a picture about reproducibility of the results, the calculations were repeated 100 times for each set of settings. The averages of $\log_{10}(\epsilon)$ (the order of relative error in variance) and $\log_{10}(1 - Q^2)$ are depicted by solid-lines and the scatters represent $\pm \sigma$ confidence intervals.

5.1. Toy 2D function

Consider a simple 2D function in which the two independent input random variables are uniformly distributed, $\mathbf{X} \sim \mathcal{U}$, with the mean values $\boldsymbol{\mu} = \{0, 0\}$ and variances $\sigma^2 = \{6, 6\}$. The design domain is thus a square $[-\sqrt{18}; \sqrt{18}]^2$. The output variable (symmetrical and highly non-normal with zero mean value and skewness) is the following transformation

$$Y = \cos\left(\frac{X_1\pi}{5}\right) \sin\left(\frac{X_2\pi}{3}\right), \quad [\sigma_Y^2 \approx 0.199\,575\,434] \quad (28)$$

which should be easily approximated by all employed methods and the whole process is easily tractable.

Non-sequential LHS was calculated for 10 increasing sample sizes in range $\langle 24, 150 \rangle$. The supposedly uniform distribution of LH-samples is compared with the proposed algorithm for sequential selection of 140 candidates (to reach the same final sample size of 150 points). The rows in Fig. 3 show selected iterations of the sequential algorithm. The color maps show the local variance density (left column), and then spatial distribution of the two components of the proposed criterion: the average variance computed using the geometrical mean with the nearest neighbor and the squared distance to the nearest neighbor. The maps on the right hand side finally present the spatial distribution of the proposed Θ criterion. A relatively large pool of LHS candidates was generated and the color maps were computed on a fine grid of coordinates. The scale of Θ in the rightmost column is proportional to the amount of variance associated with individual candidates. It can be seen that the value Θ of the best candidate also indicates how much gain in the variance accuracy can be expected by adding one point. This information can be incorporated as a kind of “stopping criterion”. The scales in Fig. 3 show that refining from $n_{\text{sim}} = 10$ to $n_{\text{sim}} = 35$ decreases the variance bites from 10^{-1} to roughly $7.5 \cdot 10^{-3}$. However, further increase in sample size (and the associated time spent on evaluation of the $g(\mathbf{x})$ function) leads only to minor decrease in Θ criterion for the best candidate and this improvement may not be deemed as worth the expense.

This reasoning is well supported by the plots of variance error in Fig. 4 (left column). One can see that from the ED with $n_{\text{sim}} \approx 25$ points, further decreases in variance error are obtained at low rates. This is emphasized by a thin vertical dashed line showing that the available polynomial basis is saturated as the maximum polynomial order is exhausted. The decision to consider is either (i) stop the algorithm: the accuracy is acceptable or computing resources are too expensive (ii) increase the maximum polynomial order p with a chance to improve the convergence rate for further size extensions. Indeed, one can see in Fig. 4 (left column, bottom plot) that after adding about 15 points to the initial design with 10 points, the maximum polynomial degree of $p = 10$ gets almost always fully exploited and the error almost stabilizes (both the error in variance estimation and also the Q^2). When the experiment is repeated with $p = 20$ (see the middle column in Fig. 4), the saturation is postponed and both errors quickly decrease until the design reaches about $n_{\text{sim}} = 120$ points. This fact documents that the adaptivity feature should also increase the polynomial degree if higher accuracy is requested.

To conclude, the proposed sequential sampling clearly outperforms standard non-sequential sampling in error in estimated variance. Although difference in the approximation error Q^2 between non-seq and sequential LHS is not significant for $p = 10$, its sequential variant leads to the most accurate estimation of variance. In case of $p = 20$, the superiority of the sequential sampling is even more significant.

For the sake of completeness, the adaptivity of p was coupled with the proposed sequential sampling (see the right column in Fig. 4). Although there exist advanced adaptive algorithms such as adaptive Coh-D opt [75], the simplest algorithm with iterative increment of p was employed for both sampling schemes, i.e. p was iteratively increased from $p = 5$ to $p = 20$ and PCE was built for a given ED and finally p yielding the lowest approximation error measured by Q^2 was selected. The obtained results show the similar behavior of the sequential technique as in case of $p = 20$, as can be expected. Note that, techniques for p -adaptivity and many other advanced sampling techniques are beyond the scope of this paper. This research is focused on the proposed sequential sampling and for the possibility of direct comparison, such type of adaptivity is excluded in the following examples in order to show purely the role of Θ criterion, though they can generally lead to more accurate approximations.

5.2. Ishigami function

Consider now a three-dimensional Ishigami function [76]. The function is strongly nonlinear, non-monotonic and presents strong interactions. We set the coefficients as in [77]. Let $\mathbf{X} \sim \mathcal{U}[-\pi, \pi]^3$ and the mathematical model

$$Y = \sin(X_1) + 7 \sin^2(X_2) + 0.1 X_3^4 \sin(X_1). \quad [\sigma_Y^2 \approx 13.844\,587\,940] \quad (29)$$

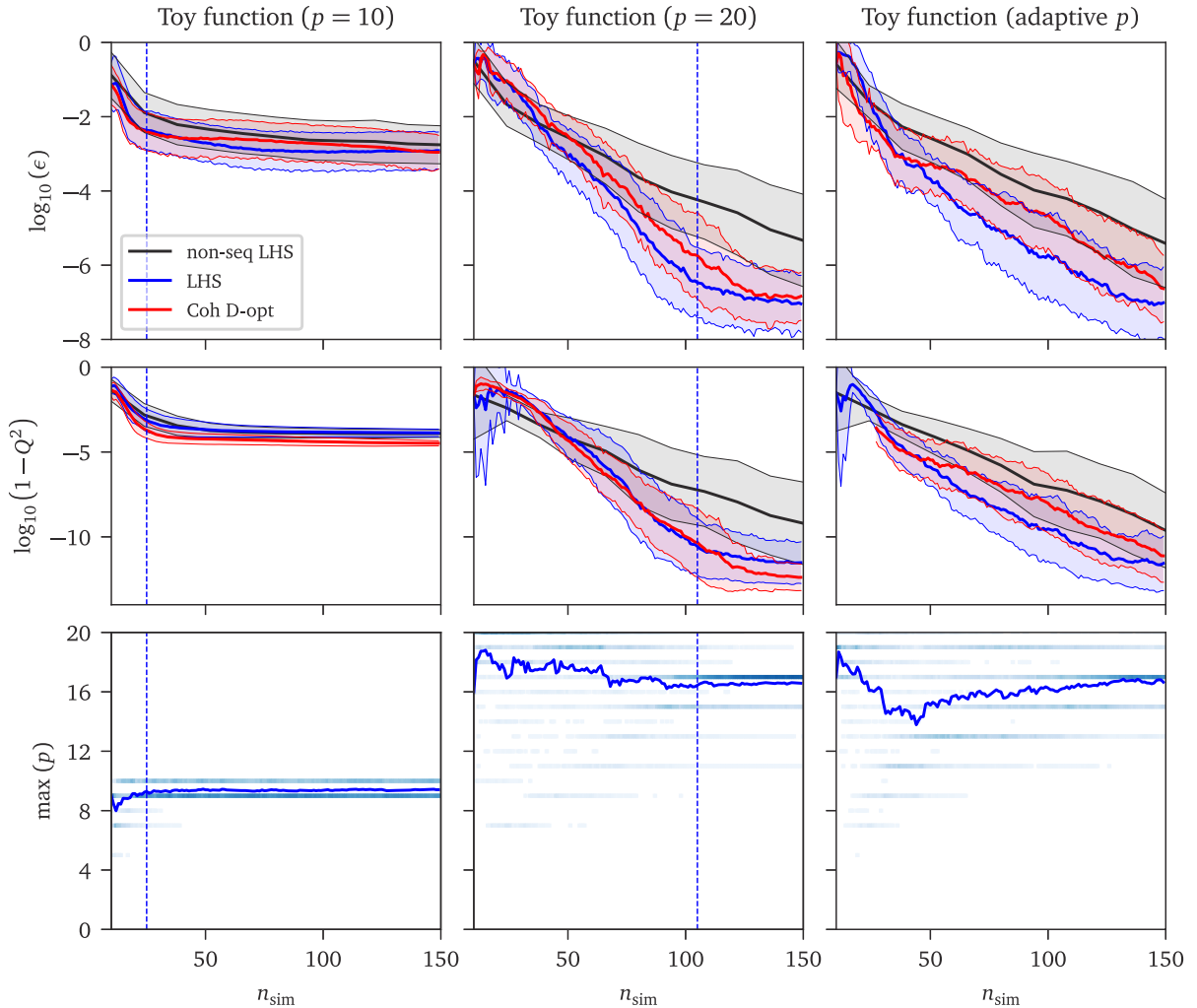


Fig. 4. Results for Toy 2D function obtained with maximum polynomial order $p = 10$ (left), $p = 20$ (middle) and adaptive p (right). The first two rows represent the accuracy measured by ϵ and Q^2 . The last row shows the mean value (solid blue line) and the empirical probability mass function (blue points) of the maximum order p in active $\mathcal{A}$ for LHS. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The Ishigami function represents a well-known benchmark function for surrogate modeling and sensitivity analysis and thus additional analysis with a very large pool size was performed in order to show additional results discussed extensively in Section 6. First of all, non-sequential PCE based on Coh-D optimal sampling was created for the direct comparison. The obtained results for standard setting are summarized in Fig. 5 (left column).

The non-sequential Coherence-D optimal ED leads to unsatisfactory results both in variance estimation and Q^2 . Although the convergence rate is lower for our purpose (low number of samples), note that it is significantly more efficient with increasing number of samples. As can be seen, the proposed sequential sampling clearly outperforms the non-seq standard approach. Moreover, the convergence rate is significantly higher, until the polynomial chaos gets saturated by reaching the maximum order $p = 10$ as can be seen in Fig. 5 (left column, bottom plot). Naturally, non-sequential technique converges to identical accuracy with increasing number of samples.

Since the Ishigami function is a low-dimensional and inexpensive example, it was also possible to perform calculation with ED pool containing a large number $n_{\text{pool}} = 5000$ of simulations and 100 repetitions in order to

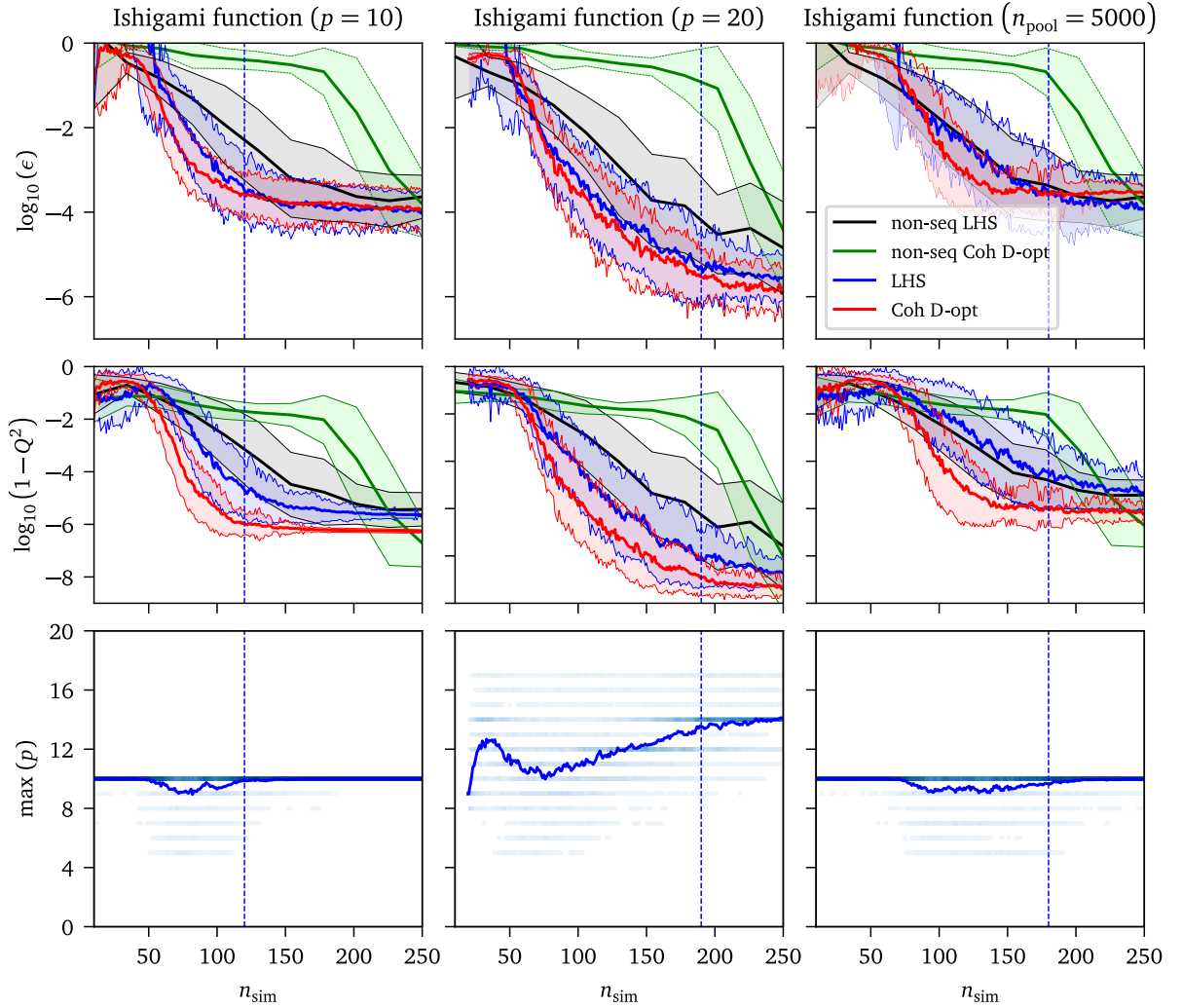


Fig. 5. Results for the Ishigami function. The first two rows represent the accuracy measured by ϵ and Q^2 . The last row shows the mean value (solid blue line) and the empirical probability mass function (blue points) of the maximum order p in active $\mathcal{A}$ for LHS. The maximum polynomial order is $p = 10$ for the left ($n_{\text{pool}} = 858$) and right columns. The pool size for the middle column is $n_{\text{pool}} = 3P = 5313$ candidate points. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

obtain statistical estimates. Note that the pool for Coh-D opt was obtained by D-optimal reduction from larger pool generated by coherence optimal sampling as in the previous examples.

The obtained results, for the case of an extremely large pool of candidates, depicted in Fig. 5 (right column) are slightly worse in comparison to the moderate size of the pool. This phenomenon is extensively discussed in Section 6. Additionally, note that the presented results are obtained for the fixed maximum polynomial order $p = 10$, which imposes a strong limit on achievable accuracy for $n_{\text{sim}} \gtrsim 120$. Similarly to the previous numerical example, allowing a higher p leads to higher accuracy of the PCE, which can be seen in Fig. 5 (middle column) using $p = 20$ (the saturation is postponed to about $n_{\text{sim}} \approx 200$).

5.3. 2D mirror line singularities

The 2D line singularities function is a mirrored version of a function used in [37]. The true surface of this function is studied for three different values of the parameter δ ; see Fig. 6. In order to document the effectiveness

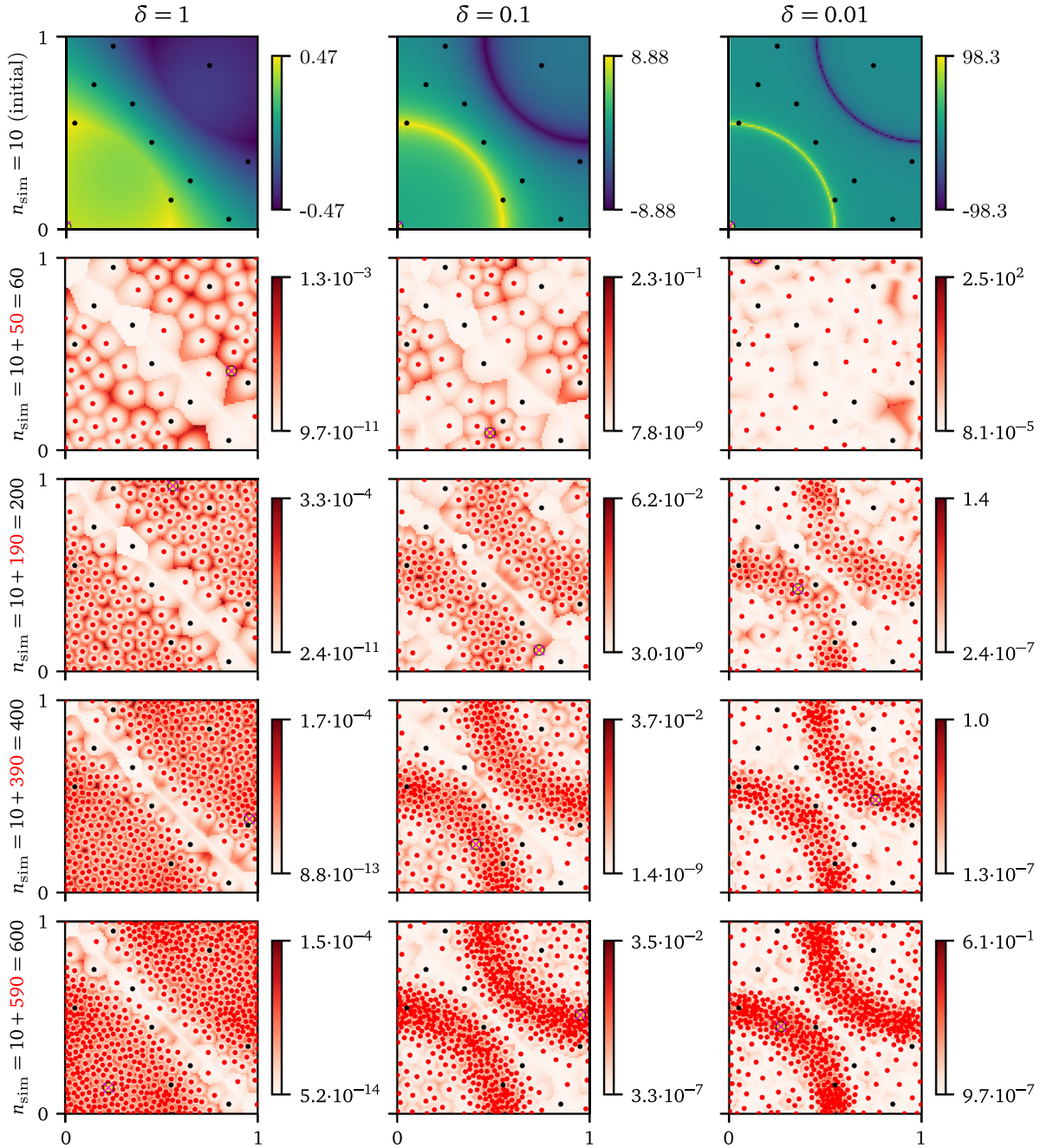


Fig. 6. 2D mirror line singularities: underlying map shows function values (the top row) and Θ criterion (others). Points represent the initial design ED (black points) and selected four iterations of algorithm adding candidates to the existing ED. Mathematical model for different parameters δ : $\delta = 1$ (left), $\delta = 0.1$ (middle), $\delta = 0.01$ (right). The maximum polynomial degree was $p = 12$ and therefore the ability of PCE to mimic a sharp singularity was limited.

of the proposed sequential sampling in the convergence plots, we selected the case of $\delta = 0.1$. Let $X \sim \mathcal{U}[0, 1]^2$ and the mathematical model be in the following form

$$Y = \frac{1}{|0.3 - X_1^2 - X_2^2| + \delta} - \frac{1}{|0.3 - (1 - X_1)^2 - (1 - X_2)^2| + \delta}. \quad [\delta = 0.1 : \sigma_Y^2 \approx 13.070477042] \quad (30)$$

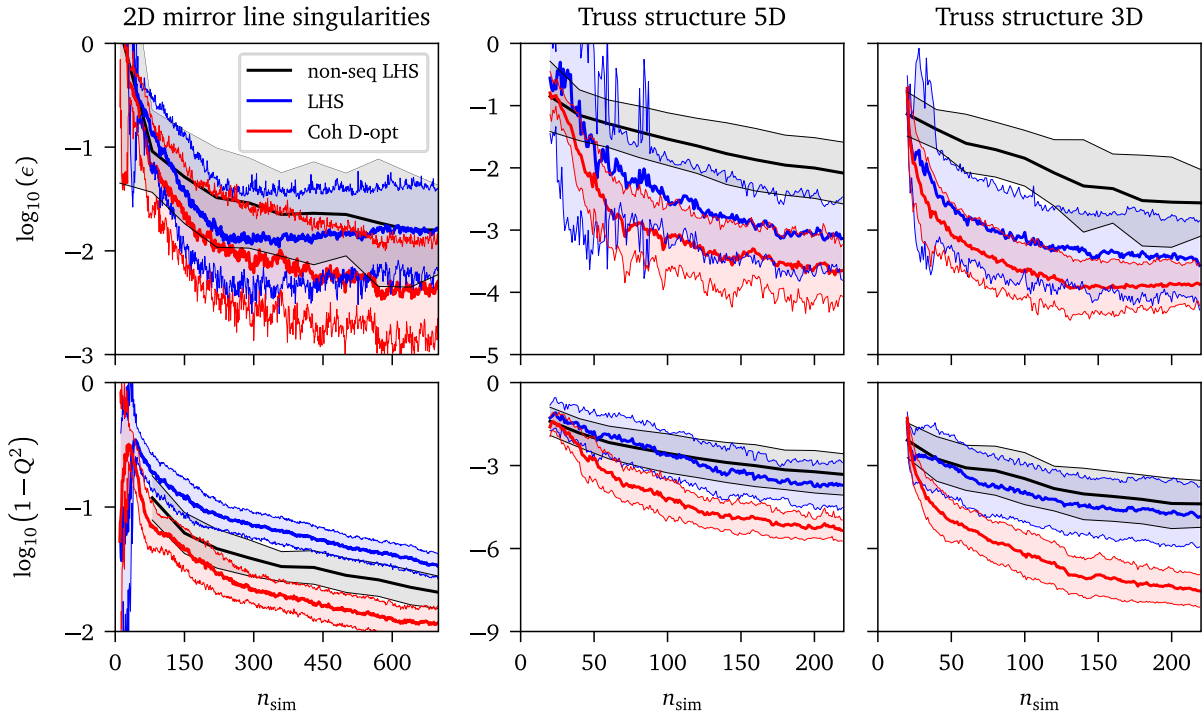


Fig. 7. Numerical results for Example 3 (left), Example 4 (middle) and Example 5 (right). The rows represent the accuracy measured via ϵ and Q^2 . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Note that, this example represents a challenging task for PCE especially for low values of δ . There are curved singularities located in a narrow vicinity of two circular arcs and thus it is crucial to identify the location of the singularity and use high-degree interacting polynomials for approximation. The values of the function range between its extremes of roughly $\pm\delta^{-1}$. We remark that the results of non-seq Coh D-opt are out of the graph range and thus this technique is not depicted in Fig. 7. Fig. 7 (top-left) shows a typical comparison of sequential and non-sequential LHS techniques. Sequential sampling is significantly better for mid-size ED, while the differences are reduced for large sample sizes when the polynomial chaos gets saturated. We remark that the high convergence rate for small to medium sample size is the practical range for which the proposed method is developed. Coh-D opt sequential strategy shows the best accuracy in Q^2 as well as in estimated variance. On the other hand, as can be seen in Fig. 7 (bottom-left) sequential LHS leads to low accuracy measured by Q^2 .

5.4. Truss structure (Hermite polynomials)

This problem involves nonuniform input variables and thus the selected set of polynomials is composed of Hermite polynomials. The design domain becomes open: $\mathbb{R}^M$. The mathematical model represents deflection of the truss structure depicted in Fig. 8. The deflection can be computed using the method of virtual work (unit load method). This method results in the following expression for the mid-span deflection

$$Y = F \left(\frac{552}{A_h E_h} + \frac{50.9117}{A_d E_d} \right). \quad [\sigma_Y^2 \approx 0.0002373] \quad (31)$$

The input random vector of the model, $\mathbf{X}$, consists of five independent random variables: the properties of the horizontal bars (Young's modulus E_h and cross-section area A_h), the properties of the diagonal bars (Young's modulus E_d and cross-section area A_d), and the magnitude of the loading forces F on the top joints. The properties of the input random variables are summarized in Table 1.

First of all, in order to construct PCE, the input random vector is transformed to standardized Gaussian random space by Nataf transformation [78,79] corresponding to Hermite germs. Further full set of polynomial basis functions

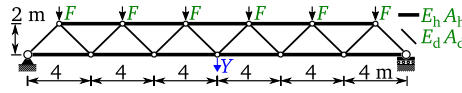


Fig. 8. Truss structure constructed from two types of bars.

Table 1

Five random variables featured in the truss example.

Variable	Distribution	Mean	Units	CoV
E_h, E_d	Log-normal	210	GPa	0.10
A_h	Log-normal	2000	mm ²	0.10
A_d	Log-normal	1000	mm ²	0.10
F	Gumbel-Max	50	kN	0.15

P is reduced by hyperbolic truncation with $q = 0.5$ according to Eq. (8). The initial ED contains $n_{\text{sim}} = 20$ realizations of X generated by LHS. We remark that Nataf transformation might increase the polynomial degree required to compute the PCE approximation of a function. To avoid the increase in nonlinearity of the problem by using the Nataf transformation, one could also compute the polynomials numerically in practical applications [80].

Obtained results can be seen in Fig. 7 (middle column). Similar to the previous example, sequential LHS leads to higher accuracy in estimated variance but the approximation error Q^2 is comparable to non-sequential LHS. The proposed sequential Coh D-opt method is clearly the most accurate and convergence rate is significantly faster both in Q^2 and variance estimation. The Coh-D sequential sampling converges to errors several orders of magnitude smaller than commonly used non-sequential LHS.

5.5. Truss structure (Hermite polynomials) — Reduced dimension

Adaptivity of the proposed algorithm is provided by the best model selection algorithm (LAR in this paper), which should be able to select the best possible set of basis functions $\mathcal{A}$. In order to examine this feature in the context of the proposed adaptive sequential sampling, the results of the previous example are compared to a manually reduced stochastic model of the previous example (Truss structure) preserving identical mathematical model as follows [81]:

$$Y = F \left(\frac{552}{\mathcal{A}_h} + \frac{50.9117}{\mathcal{A}_d} \right), \quad [\sigma_Y^2 \approx 0.0002373] \quad (32)$$

where $\mathcal{A}_h$ is a Log-normal random variable with mean of 420 MN and $\mathcal{A}_d$ is a Log-normal random variable with mean of 210 MN, CoV of both random variables equals 0.14177. The results obtained from the sequential sampling are summarized in Fig. 7 (right). As can be seen, the final accuracy of this example measured by relative error in variance ϵ is similar to the 5D formulation of the $g(X)$ when the size of ED reaches the final $n_{\text{sim}} = 220$, although the convergence of the reduced model to this value is significantly faster for lower n_{sim} . This is in compliance with the theoretical behavior of the model selection algorithm, which becomes more efficient with greater samples size. The faster convergence also affects the final accuracy of Coh D-opt measured by Q^2 , which is able to converge to lower values for the final n_{sim} .

6. Discussion

6.1. Optimal pool size and the maximum polynomial order p

As already briefly mentioned above, the selection of fixed maximum polynomial order p may impose a lower bound on achievable accuracy. This phenomenon was visible in the first “Toy” example (see Fig. 4) as well as for the “Ishigami function”, see Fig. 5. In these examples, the initial guess on $p = 10$ was found insufficient and selection of $p = 20$ allowed for much higher accuracy of the surrogate. This is documented by examination of the set $\mathcal{A}$, specifically maximum p in the active set of basis functions $\mathcal{A}$ chosen by LAR depicted by blue color in the bottom rows of Figs. 4 and 5. It can be seen, that convergence rate is very low once the maximum order p in

active $\mathcal{A}$ achieves its prescribed maximum of $p = 10$ since there are no additional possibilities for improvement of the approximation (the space of basis functions is out of suitable candidates). Therefore, if one extends the space of basis functions by sufficiently incrementing p , the decrease in error does not stop and ϵ for maximum n_{sim} is significantly lower. Unfortunately, the ideal p is not known a priori and thus one should adaptively increase it with growing active ED. However, as discussed already in [2], it is possible to employ an adaptive selection of the best p for each iteration. Such a feature might play a significant role in some cases. Such approach was deliberately not employed in this paper because the influence of the proposed sampling scheme might not be clearly separable from the model selection.

The proposed sequential technique achieved higher accuracy of estimated variance for lower number of samples in comparison to non-sequential sampling in all presented numerical examples. However, as can be seen from numerical results of the Ishigami function, increasing the pool size does not generally lead to higher accuracy or faster convergence rate. In fact, a very large number of candidates might cause slower convergence (comparable to non-sequential sampling). After detailed examination of this example, one can see the different structure of $\mathcal{A}$ for the large set of candidates depicted in Fig. 5 (rightmost column). During the initial phase of the adaptive sequential sampling, LAR algorithm typically selects high-order polynomials since there is not enough information about a mathematical model, which leads to *overfitting*. Further increasing the sample size enables the adaptive algorithm to identify more appropriate low-order basis functions ($n_{\text{sim}} \approx 50$ for Ishigami function). Sequential sampling in the following steps of algorithm selects the candidates with respect to the current $\mathcal{A}$ and exploration aspect. Note that high-order basis functions are selected by adaptive algorithm once the new regions associated to high local variance are discovered (peaks of the given mathematical model). However, such regions are not preferred by the selection criterion since low-order basis functions ignore the currently unknown extremes. Therefore, the convergence rate of the adaptive sequential sampling can be significantly affected by the number of candidates, since if there are no candidates in regions favored by low-order basis functions, the exploration part of the proposed criterion could investigate new functional extremes and adapt $\mathcal{A}$ to a set of high-order basis functions. This phenomenon is illustrated by blue color in numerical results in Fig. 5 (left). In the first case with $n_{\text{pool}} = 3P$ (and $p = 10$), one can see a fast convergence for n_{sim} ranging between 50 and 150 (associated to lower maximum p in active $\mathcal{A}$) until the space of basis functions is out of suitable candidates as described in the paragraph above. On the other hand, for the second case with $n_{\text{pool}} = 5\,000$ (and $p = 10$ again; see Fig. 5 right), an active $\mathcal{A}$ contains low-order basis functions for a range of significantly greater n_{sim} , which leads to a slower convergence of accuracy ultimately leading to the identical final error.

In summary, the achievable accuracy of the adaptive sequential sampling is limited by the maximum polynomial order p used. Simply, the flexibility of the polynomial approximation may not be sufficient to approximate the original function at a given precision level. In practical applications, a sufficient polynomial order is not known a priori and therefore p should be adaptively increased with increasing n_{sim} in order to achieve the best performance of the proposed algorithm. On the other hand, the convergence rate is significantly affected by n_{pool} and extremely large pool leads to slow convergence, since a selection of high-order basis functions is postponed to higher n_{sim} . Therefore the heuristic rule $n_{\text{pool}} = 3P$ is recommended for practical applications.

6.2. High dimensions

What remains an open question is the behavior of the proposed criterion in high dimensions. In particular, we need to understand the effect of the exploration part $l_{c,s}^M$ in Eq. (24). Generally, in high-dimensional space with independent and identically distributed (iid) components, the Minkowski distance of order $\mathcal{P} > 0$ (sometimes referred to as the $\mathcal{P}$ -norm) concentrates, i.e. the coefficient of variation of the norm decreases with increasing dimension, M . This effect on $\mathcal{P}$ -norm of letting M go large is well known in the computational learning literature. A discussion in the context of iid Gaussian distribution can be found e.g. in [82]. The asymptotic behavior of $\mathcal{P}$ -norm and its convergence rates have been studied in [83].

The role of the distance term must be discussed especially in the case of uniform distribution of the germ which is defined over a hypercube. It is known that Euclidean distances (a special case of Minkowski distance with $\mathcal{P} = 2$) of any pair of points inside a hypercube tend to concentrate around its mean value when the dimension is high. It is known that the standard deviation of the Euclidean distance between any two randomly picked points stays approximately constant with increasing dimension while the mean value keeps growing proportionally to

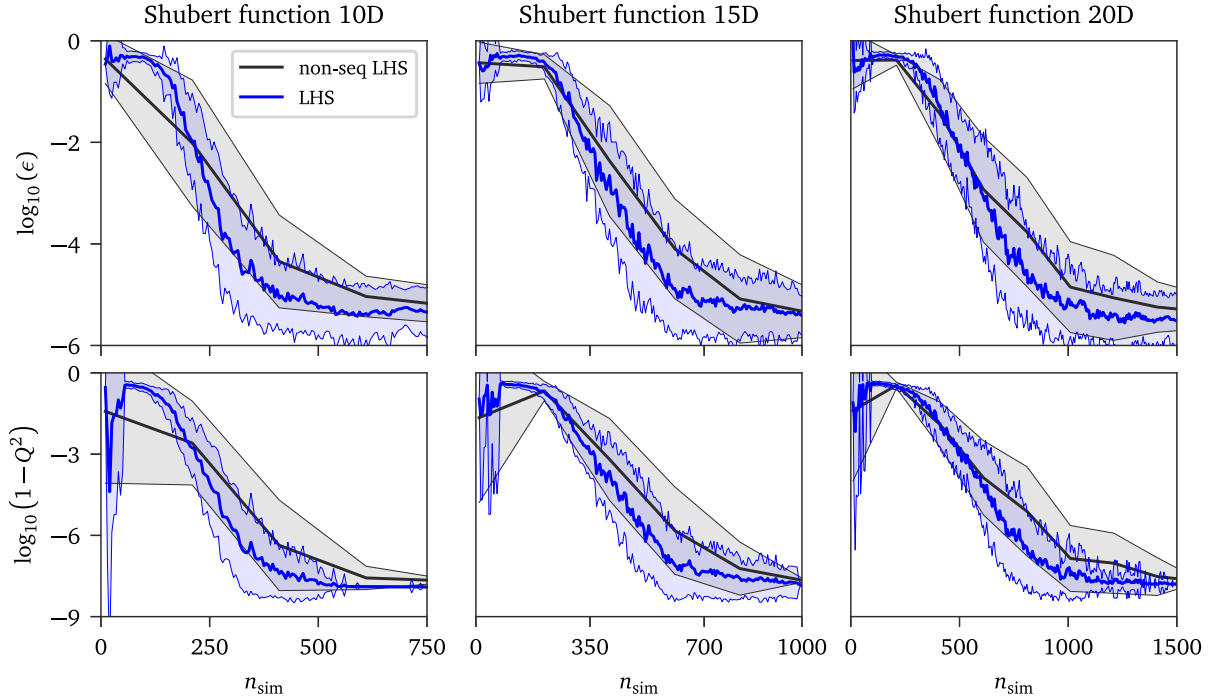


Fig. 9. Results for the Shubert function in 10 (left), 15 (middle) and 20 (right) dimensions. The first row represents the accuracy of estimated variance measured by ϵ and the second row shows the approximation accuracy measured by Q^2 .

$\sqrt{M}$ [84,85]. Therefore, the coefficient of variation of a random distance is asymptotically proportional to $1/\sqrt{M}$. In such a case the distance contrast decreases and it is said that the distances concentrate [86,87]. The same holds also for the squared Euclidean distance l^2 between points picked at random: the squared distance has Gaussian distribution with the mean value of $M/6$ and variance $7M/180$. In order to obtain the hypervolume of a region in between a pair of points, the squared distance must be raised to $M/2$. We have found that the coefficient of variation of such volume quickly increases as the problem dimension grows high. Therefore, there is no problem with insufficient contrast as the proposed Θ criterion sufficiently varies even in domains of high dimension.

Although it will be necessary to perform extensive study with advanced sampling schemes in order to investigate the efficiency of the proposed criterion in high dimensions, the general behavior of the Θ criterion is demonstrated in the following simple example with input random vector $X \sim \mathcal{U}[-1, 1]^M$ — Shubert function No. 4 [88]:

$$Y = \sum_{i=1}^M \sum_{j=1}^5 j \cos((j+1)X_i + j). \quad [\sigma_Y^2 \approx M \cdot 45.266\,621\,664] \quad (33)$$

This function can be easily utilized as a benchmark for an arbitrary number of input random variables; in our study we use $M = \{10, 15, 20\}$. The maximum polynomial order $p = 12$ and hyperbolic truncation parameter $q = 0.3$ were used for construction of surrogate model, and the initial ED contains $n_{\text{sim}} = 10$ realizations of X generated by LHS. Since the hyperbolic truncation significantly reduces the P in high dimensions, we increased the pool size to $n_{\text{pool}} = 5P$. Note that, this study is focused only on stability of the proposed Θ criterion in higher dimensions and thus only the sequential and the non-sequential LHS were employed for direct comparison. From the obtained results depicted in Fig. 9, it can be seen that the proposed criterion enabling one-by-one extension of ED achieves higher accuracy in comparison to non-sequential approach independently of dimension M and thus the theoretical discussion in the previous paragraph was presented also numerically. Naturally, the absolute value of accuracy is highly dependent on the investigated function and on the particular sampling scheme coupled with Θ criterion, however this study demonstrates the applicability of the proposed criterion in high dimensions.

6.3. Further work

Since the DoE is usually not a bottleneck in probabilistic analysis and sequential sampling offers several advantages in comparison to non-sequential sampling, it can be recommended to employ the proposed algorithm for the construction of PCE despite higher computational requirements of repetitive DoE in each step. Although the choice of the sampling technique coupled with the proposed criterion for the sequential selection of the best candidate is arbitrary, it can be seen from the numerical examples, that LHS is efficient for low dimensional mathematical models. On the other hand, coherence D-optimal sampling achieved significantly higher accuracy in more complex examples. Therefore further work will be focused on an improvement of effectiveness by using advanced optimized space-filling designs for the generating of a pool of candidates and a comparative study of existing advanced sampling techniques coupled with the proposed criterion will be performed. Employment of designs generated from the target distribution that additionally avoids clustering [17,81,89] or empty regions [60] while maintaining true statistical homogeneity via periodic distance-based criteria has the potential to further improve the effectiveness of the proposed method, especially in high-dimensional space. Moreover, it was shown that fixed p represents a significant limitation for the proposed method and thus further work will be also focused on adaptive basis strategies [75,90], which have the potential to dramatically improve the final accuracy of PCE and solve the problem with LAR and with a large size of the pool of candidates.

7. Conclusion

A novel adaptive sequential sampling technique for accurate and efficient construction of a non-intrusive PCE was proposed in this paper and its performance was validated on several numerical examples of increasing complexity and dimensionality. The proposed technique selects the best candidate sample from a large pool maintaining the balance between exploration of the design domain and exploitation of the current characteristics of the PCE. The criterion driving the selection of the best candidate was successfully coupled with LHS and Coh-D optimal sampling and both variants were used in numerical examples. From the obtained results, it can be concluded that the proposed technique leads to higher accuracy of the constructed PCE in comparison to non-sequential sampling. The difference in accuracy between sequential and non-sequential sampling is especially significant for a low-size ED. However, it can be expected that the accuracy of sequential sampling converges to identical results as a non-sequential sampling for a very large ED in which the PCE is saturated. Comparing both sequential sampling techniques, superior performance was achieved by sequential adaptive Coh-D optimal sampling due to its adaptivity of the candidate sample in each iteration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge financial support provided by the Ministry of Education, Youth and Sports of the Czech Republic under Project No. LTAUSA19058. Additionally, the first author is also supported by the Ministry of Education, Youth and Sports of the Czech Republic under Project No. FAST-J-21-7209.

References

- [1] N. Wiener, The homogeneous chaos, *Amer. J. Math.* 60 (4) (1938) 897–936, <http://dx.doi.org/10.2307/2371268>.
- [2] G. Blatman, B. Sudret, Adaptive sparse polynomial chaos expansion based on least angle regression, *J. Comput. Phys.* 230 (6) (2011) 2345–2367, <http://dx.doi.org/10.1016/j.jcp.2010.12.021>.
- [3] R.G. Ghanem, P.D. Spanos, *Stochastic Finite Elements: A Spectral Approach*, Springer, New York, 1991, <http://dx.doi.org/10.1007/978-1-4612-3094-6>.
- [4] T. Crestaux, O.L. Maître, J.-M. Martinez, Polynomial chaos expansion for sensitivity analysis, *Reliab. Eng. Syst. Saf.* 94 (7) (2009) 1161–1172, <http://dx.doi.org/10.1016/j.res.2008.10.008>.
- [5] B. Sudret, Global sensitivity analysis using polynomial chaos expansions, *Reliab. Eng. Syst. Saf.* 93 (7) (2008) 964–979, <http://dx.doi.org/10.1016/j.res.2007.04.002>.

- [6] N.-Z. Chen, C. Guedes Soares, Spectral stochastic finite element analysis for laminated composite plates, *Comput. Methods Appl. Mech. Eng.* 197 (51) (2008) 4830–4839, <http://dx.doi.org/10.1016/j.cma.2008.07.003>.
- [7] L. Novak, D. Novak, Surrogate modelling in the stochastic analysis of concrete girders failing in shear, in: *Proc. of the Fib Symposium 2019: Concrete - Innovations in Materials, Design and Structures*, 2019, pp. 1741–1747.
- [8] M. Chatzimanolakis, K.-D. Kantarakias, V. Asouti, K. Giannakoglou, A painless intrusive polynomial chaos method with RANS-based applications, *Comput. Methods Appl. Mech. Eng.* 348 (2019) 207–221, <http://dx.doi.org/10.1016/j.cma.2019.01.018>.
- [9] G.T. Buzzard, Global sensitivity analysis using sparse grid interpolation and polynomial chaos, *Reliab. Eng. Syst. Saf.* 107 (2012) 82–89, <http://dx.doi.org/10.1016/j.jress.2011.07.011>, SAMO 2010.
- [10] Z. Perkó, L. Gilli, D. Lathouwers, J.L. Kloosterman, Grid and basis adaptive polynomial chaos techniques for sensitivity and uncertainty analysis, *J. Comput. Phys.* 260 (2014) 54–84, <http://dx.doi.org/10.1016/j.jcp.2013.12.025>.
- [11] A. Cohen, G. Migliorati, Optimal weighted least-squares methods, *SMAI J. Comput. Math.* 3 (2017) 181–203, <http://dx.doi.org/10.5802/smai-jcm.24>.
- [12] A.C. Narayan, J. Jakeman, T. Zhou, A christoffel function weighted least squares algorithm for collocation approximations, *Math. Comp.* 86 (2017) 1913–1947, <http://dx.doi.org/10.1090/mcom/3192>.
- [13] N. Lüthen, S. Marelli, B. Sudret, Sparse polynomial chaos expansions: Literature survey and benchmark, *SIAM/ASA J. Uncertain. Quantif.* 9 (2) (2021) 593–649, <http://dx.doi.org/10.1137/20M1315774>.
- [14] W. Conover, On a better method for selecting input variables, 1975, unpublished Los Alamos National Laboratories manuscript, reproduced as Appendix A of “Latin Hypercube Sampling and the Propagation of Uncertainty in Analyses of Complex Systems” by J.C. Helton and F.J. Davis, Sandia National Laboratories report SAND2001-0417, printed November 2002. URL <https://prod-ng.sandia.gov/techlib-noauth/access-control.cgi/2001/010417.pdf>.
- [15] M. Johnson, L. Moore, D. Ylvisaker, Minimax and maximin distance designs, *J. Statist. Plann. Inference* 2 (26) (1990) 131–148, [http://dx.doi.org/10.1016/0378-3758\(90\)90122-B](http://dx.doi.org/10.1016/0378-3758(90)90122-B).
- [16] M.D. Morris, T.J. Mitchell, Exploratory designs for computational experiments, *J. Statist. Plann. Inference* 43 (3) (1995) 381–402, [http://dx.doi.org/10.1016/0378-3758\(94\)00035-T](http://dx.doi.org/10.1016/0378-3758(94)00035-T).
- [17] M. Vořechovský, J. Eliáš, Modification of the Maximin and ϕ_p (phi) criteria to achieve statistically uniform distribution of sampling points, *Technometrics* 62 (3) (2020) 371–386, <http://dx.doi.org/10.1080/00401706.2019.1639550>.
- [18] J. Halton, On the efficiency of certain quasi-random sequences of points in evaluating multi-dimensional integrals, *Numer. Math.* 2 (1) (1960) 84–90, <http://dx.doi.org/10.1007/BF01386213>.
- [19] H. Faure, C. Lemieux, Generalized Halton sequences in 2008: A comparative study, *ACM Trans. Model. Comput. Simul.* 19 (4) (2009) 15:1–15:31, <http://dx.doi.org/10.1145/1596519.1596520>.
- [20] I. Sobol’, On the distribution of points in a cube and the approximate evaluation of integrals, *USSR Comput. Math. Math. Phys.* 7 (1967) [http://dx.doi.org/10.1016/0041-5553\(67\)90144-9](http://dx.doi.org/10.1016/0041-5553(67)90144-9).
- [21] I.M. Sobol’, Uniformly distributed sequences with an additional uniform property, *USSR Comput. Math. Math. Phys.* 16 (5) (1976) 236–242, [http://dx.doi.org/10.1016/0041-5553\(76\)90154-3](http://dx.doi.org/10.1016/0041-5553(76)90154-3), Short communication.
- [22] H. Niederreiter, Point sets and sequences with small discrepancy, *Monatsh. Math.* 104 (4) (1987) 273–337, <http://dx.doi.org/10.1007/BF01294651>.
- [23] H. Niederreiter, Low-discrepancy and low-dispersion sequences, *J. Number Theory* 30 (1) (1988) 51–70, [http://dx.doi.org/10.1016/0022-314X\(88\)90025-X](http://dx.doi.org/10.1016/0022-314X(88)90025-X).
- [24] H. Niederreiter, Random Number Generation and Quasi-Monte Carlo Methods, in: *CBMS-NSF Regional Conference Series in Applied Mathematics*, Society for Industrial and Applied Mathematics (SIAM), Philadelphia, Pennsylvania, 1992, <http://dx.doi.org/10.1137/1.9781611970081>.
- [25] H. Faure, Discrepances de suites associées à un système de numération (en dimension un) [Discrepancy of sequences associated with a number system (in dimension one)], *Bull. Soc. Math. France* 109 (1981).
- [26] S. Tezuka, Uniform Random Numbers: Theory and Practice, in: *The Springer International Series in Engineering and Computer Science* 315, Springer, Boston, MA, 1995, <http://dx.doi.org/10.1007/978-1-4615-2317-8>.
- [27] J. Hampton, A. Doostan, Compressive sampling of polynomial chaos expansions: Convergence analysis and sampling strategies, *J. Comput. Phys.* 280 (2015) 363–386, <http://dx.doi.org/10.1016/j.jcp.2014.09.019>.
- [28] A. Cohen, G. Migliorati, Optimal weighted least-squares methods, *SMAI J. Comput. Math.* 3 (2017) 181–203, <http://dx.doi.org/10.5802/smai-jcm.24>.
- [29] B. Echar, N. Gayton, M. Lemaire, AK-MCS: An active learning reliability method combining Kriging and Monte Carlo simulation, *Struct. Saf.* 33 (2) (2011) 145–154, <http://dx.doi.org/10.1016/j.strusafe.2011.01.002>.
- [30] L. Shi, B. Sun, D.S. Ibrahim, An active learning reliability method with multiple kernel functions based on radial basis function, *Struct. Multidiscip. Optim.* 60 (1) (2019) 211–229, <http://dx.doi.org/10.1007/s00158-019-02210-0>.
- [31] X. Yang, X. Cheng, Active learning method combining Kriging model and multimodal-optimization-based importance sampling for the estimation of small failure probability, *Internat. J. Numer. Methods Engrg.* 121 (21) (2020) 4843–4864, <http://dx.doi.org/10.1002/nme.6495>.
- [32] S. Marelli, B. Sudret, An active-learning algorithm that combines sparse polynomial chaos expansions and bootstrap for structural reliability analysis, *Struct. Saf.* 75 (2018) 67–74, <http://dx.doi.org/10.1016/j.strusafe.2018.06.003>.
- [33] Y. Zhou, Z. Lu, W. Yun, Active sparse polynomial chaos expansion for system reliability analysis, *Reliab. Eng. Syst. Saf.* 202 (2020) 107025, <http://dx.doi.org/10.1016/j.jress.2020.107025>.
- [34] K. Cheng, Z. Lu, Active learning polynomial chaos expansion for reliability analysis by maximizing expected indicator function prediction error, *Internat. J. Numer. Methods Engrg.* 121 (14) (2020) 3159–3177, <http://dx.doi.org/10.1002/nme.6351>.

- [35] N. Fajraoui, S. Marelli, B. Sudret, Sequential design of experiment for sparse polynomial chaos expansions, *SIAM/ASA J. Uncertain. Quantif.* 5 (1) (2017) 1061–1085, <http://dx.doi.org/10.1137/16m1103488>.
- [36] M. Thapa, S.B. Mulani, R.W. Walters, Adaptive weighted least-squares polynomial chaos expansion with basis adaptivity and sequential adaptive sampling, *Comput. Methods Appl. Mech. Eng.* 360 (2020) 112759, <http://dx.doi.org/10.1016/j.cma.2019.112759>.
- [37] M.D. Shields, Adaptive Monte Carlo analysis for strongly nonlinear stochastic systems, *Reliab. Eng. Syst. Saf.* 175 (2018) 207–224, <http://dx.doi.org/10.1016/j.res.2018.03.018>.
- [38] Y. Zhou, Z. Lu, K. Cheng, C. Ling, An efficient and robust adaptive sampling method for polynomial chaos expansion in sparse Bayesian learning framework, *Comput. Methods Appl. Mech. Engrg.* 352 (2019) 654–674, <http://dx.doi.org/10.1016/j.cma.2019.04.046>.
- [39] S. Marelli, B. Sudret, UQLab: A framework for uncertainty quantification in matlab, in: *Vulnerability, Uncertainty, and Risk*, 2014, pp. 2554–2563, <http://dx.doi.org/10.1061/9780784413609.257>.
- [40] E. Patelli, S. Tolo, H. George-Williams, J. Sadeghi, R. Rocchetta, M. de Angelis, M. Broggi, OpenCossan 2.0: an efficient computational toolbox for risk, reliability and resilience analysis, in: *Proceedings of the Joint ICVRAM ISUMA UNCERTAINTIES Conference*, 2018, pp. 1–8, URL <http://icvramisuma2018.org/cd/web/PDF/ICVRAMISUMA2018-0022.PDF>.
- [41] L. Novak, D. Novak, Polynomial chaos expansion for surrogate modelling: Theory and software, *Beton- Stahlbetonbau* 113 (2018) 27–32, <http://dx.doi.org/10.1002/best.201800048>.
- [42] J. Feinberg, H.P. Langtangen, Chaospy: An open source tool for designing methods of uncertainty quantification, *J. Comput. Sci.* 11 (2015) 46–57, <http://dx.doi.org/10.1016/j.jocs.2015.08.008>.
- [43] A. Olivier, D. Giovanis, B. Aakash, M. Chauhan, L. Vandanapu, M.D. Shields, UQpy: A general purpose Python package and development environment for uncertainty quantification, *J. Comput. Sci.* 47 (2020) 101204.
- [44] D. Xiu, G.E. Karniadakis, The Wiener–Askey polynomial chaos for stochastic differential equations, *SIAM J. Sci. Comput.* 24 (2) (2002) 619–644, <http://dx.doi.org/10.1137/s1064827501387826>.
- [45] B. Efron, T. Hastie, I. Johnstone, R. Tibshirani, Least angle regression, *Ann. Statist.* 32 (2) (2004) 407–451, <http://dx.doi.org/10.2307/3448465>.
- [46] J.A. Tropp, A.C. Gilbert, Signal recovery from random measurements via orthogonal matching pursuit, *IEEE Trans. Inf. Theory* 53 (12) (2007) 4655–4666, <http://dx.doi.org/10.1109/tit.2007.909108>.
- [47] S. Ji, Y. Xue, L. Carin, Bayesian compressive sensing, *IEEE Trans. Signal Process.* 56 (6) (2008) 2346–2356, <http://dx.doi.org/10.1109/TSP.2007.914345>.
- [48] G. Blatman, B. Sudret, An adaptive algorithm to build up sparse polynomial chaos expansions for stochastic finite element analysis, *Probab. Eng. Mech.* 25 (2) (2010) 183–197, <http://dx.doi.org/10.1016/j.probengmech.2009.10.003>.
- [49] J.F. Koksma, Een algemeene stelling uit de theorie der gelijkmatige verdeling modulo 1, *Math. B* 11 (1942/1943) 7–11.
- [50] K.-T. Fang, Y. Wang, *Number-Theoretic Methods in Statistics*, first ed., Chapman and Hall/CRC, London; New York, 1993.
- [51] F.J. Hickernell, A generalized discrepancy and quadrature error bound, *Math. Comp.* 67 (221) (1998) 299–322, <http://dx.doi.org/10.1090/S0025-5718-98-00894-1>.
- [52] F.J. Hickernell, Lattice rules: How well do they measure up? in: P. Hellekalek, G. Larcher (Eds.), *Random and Quasi-Random Point Sets*, Springer New York, New York, NY, 1998, pp. 109–166, http://dx.doi.org/10.1007/978-1-4612-1702-2_3, Ch. Lattice Rules: How Well Do They Measure Up?.
- [53] J. Dick, F. Pillichshammer, *Digital Nets and Sequences: Discrepancy Theory and Quasi-Monte Carlo Integration*, Cambridge University Press, 2010, <http://dx.doi.org/10.1017/cbo9780511761188>.
- [54] M.D. McKay, W.J. Conover, R.J. Beckman, A comparison of three methods for selecting values of input variables in the analysis of output from a computer code, *Technometrics* 21 (1979) 239–245, <http://dx.doi.org/10.1080/00401706.1979.10489755>.
- [55] M. Stein, Large sample properties of simulations using Latin hypercube sampling, *Technometrics* 29 (2) (1987) 143–151, <http://dx.doi.org/10.1080/00401706.1987.10488205>.
- [56] A.B. Owen, A central limit theorem for Latin hypercube sampling, *J. R. Stat. Soc. Ser. B Stat. Methodol.* 54 (2) (1992) 541–551, <http://dx.doi.org/10.2307/2346140>.
- [57] A.B. Owen, Controlling correlations in latin hypercube samples, *J. Amer. Statist. Assoc. (Theory Methods)* 89 (428) (1994) 1517–1522.
- [58] M.D. Shields, J. Zhang, The generalization of Latin hypercube sampling, *Reliab. Eng. Syst. Saf.* 148 (2016) 96–108, <http://dx.doi.org/10.1016/j.res.2015.12.002>.
- [59] L. Pronzato, Minimax and maximin space-filling designs: some properties and methods for construction, *J. Soc. Fr. Stat.* 158 (1) (2017) 7–36.
- [60] J. Eliáš, M. Vořechovský, V. Sadílek, Periodic version of the minimax distance criterion for Monte Carlo integration, *Adv. Eng. Softw.* 149 (2020) 102900, <http://dx.doi.org/10.1016/j.advengsoft.2020.102900>.
- [61] S.-K. Choi, R.V. Grandhi, R.A. Canfield, C.L. Pettit, Polynomial chaos expansion with latin hypercube sampling for estimating response variability, *AIAA J.* 42 (6) (2004) 1191–1198, <http://dx.doi.org/10.2514/1.2220>.
- [62] M. Hadigol, A. Doostan, Least squares polynomial chaos expansion: A review of sampling strategies, *Comput. Methods Appl. Mech. Eng.* 332 (2018) 382–407, <http://dx.doi.org/10.1016/j.cma.2017.12.019>.
- [63] E.J. Candes, Y. Plan, A probabilistic and RIPless theory of compressed sensing, *IEEE Trans. Inf. Theory* 57 (11) (2011) 7235–7254, <http://dx.doi.org/10.1109/tit.2011.2161794>.
- [64] J.D. Jakeman, A. Narayan, T. Zhou, A generalized sampling and preconditioning scheme for sparse approximation of polynomial chaos expansions, *SIAM J. Sci. Comput.* 39 (3) (2017) A1114–A1144, <http://dx.doi.org/10.1137/16M1063885>.
- [65] J. Hampton, A. Doostan, Coherence motivated sampling and convergence analysis of least squares polynomial chaos regression, *Comput. Methods Appl. Mech. Eng.* 290 (2015) 73–97, <http://dx.doi.org/10.1016/j.cma.2015.02.006>.
- [66] A.J. Miller, N.-K. Nguyen, Algorithm AS 295: A fedorov exchange algorithm for D-optimal design, *J. R. Stat. Soc. Ser. C (Appl. Stat.)* 43 (4) (1994) 669–677, <http://dx.doi.org/10.2307/2986264>.

- [67] O. Dykstra, The augmentation of experimental data to maximize $[X'X]$, *Technometrics* 13 (3) (1971) 682–688, <http://dx.doi.org/10.1080/00401706.1971.10488830>.
- [68] P. Diaz, A. Doostan, J. Hampton, Sparse polynomial chaos expansions via compressed sensing and D-optimal design, *Comput. Methods Appl. Mech. Eng.* 336 (2018) 640–666, <http://dx.doi.org/10.1016/j.cma.2018.03.020>.
- [69] Z. Wu, D. Wang, P.N. Okolo, K. Zhao, W. Zhang, Efficient space-filling and near-orthogonality sequential latin hypercube for computer experiments, *Comput. Methods Appl. Mech. Eng.* 324 (2017) 348–365, <http://dx.doi.org/10.1016/j.cma.2017.05.020>.
- [70] C. Tong, Refinement strategies for stratified sampling methods, *Reliab. Eng. Syst. Saf.* 91 (10) (2006) 1257–1265, <http://dx.doi.org/10.1016/j.res.2005.11.027>.
- [71] M. Vořechovský, Hierarchical refinement of Latin Hypercube Samples, *Comput.-Aided Civ. Infrastruct. Eng.* 30 (5) (2015) 394–411, <http://dx.doi.org/10.1111/mice.12088>.
- [72] M. Shields, Refined latinized stratified sampling: a robust sequential sample size extension methodology for high-dimensional latin hypercube and stratified designs, *Int. J. Uncertain. Quantif.* 6 (2016) 79–97.
- [73] M.D. Shields, K. Teffera, A. Hapji, R.P. Daddazio, Refined stratified sampling for efficient Monte Carlo based uncertainty quantification, *Reliab. Eng. Syst. Saf.* 142 (2015) 310–325.
- [74] J.J. Borkowski, E.S. Valeroso, Comparison of design optimality criteria of reduced models for response surface designs in the hypercube, *Technometrics* 43 (4) (2001) 468–477, <http://dx.doi.org/10.1198/00401700152672564>.
- [75] J. Hampton, A. Doostan, Basis adaptive sample efficient polynomial chaos (BASE-PC), *J. Comput. Phys.* 371 (2018) 20–49, <http://dx.doi.org/10.1016/j.jcp.2018.03.035>.
- [76] T. Ishigami, An importance quantification technique in uncertainty analysis for computer models, in: *Proceedings. First International Symposium on Uncertainty Modeling and Analysis*, IEEE Comput. Soc. Press, 1990, pp. 398–403, <http://dx.doi.org/10.1109/isuma.1990.151285>.
- [77] A. Marrel, B. Iooss, B. Laurent, O. Roustant, Calculations of Sobol indices for the Gaussian process metamodel, *Reliab. Eng. Syst. Saf.* 94 (3) (2009) 742–751, <http://dx.doi.org/10.1016/j.res.2008.07.008>.
- [78] A. Nataf, Détermination des distributions de probabilité dont les marges sont données, *C. R. Acad. Sci.* 225 (1962) 42–43.
- [79] R. Lebrun, A. Dutfoy, An innovating analysis of the Nataf transformation from the copula viewpoint, *Probab. Eng. Mech.* 24 (3) (2009) 312–320, <http://dx.doi.org/10.1016/j.probangmech.2008.08.001>.
- [80] J.D. Jakeman, F. Franzelin, A. Narayan, M. Eldred, D. Pflüger, Polynomial chaos expansions for dependent random variables, *Comput. Methods Appl. Mech. Engrg.* 351 (2019) 643–666, <http://dx.doi.org/10.1016/j.cma.2019.03.049>.
- [81] M. Vořechovský, J. Mašek, J. Eliáš, Distance-based optimal sampling in a hypercube: Analogies to N-body systems, *Adv. Eng. Softw.* 137 (2019) 102709, <http://dx.doi.org/10.1016/j.advengsoft.2019.102709>.
- [82] L. Katafygiotis, K. Zuev, Geometric insight into the challenges of solving high-dimensional reliability problems, *Probab. Eng. Mech.* 23 (2–3) (2008) 208–218, <http://dx.doi.org/10.1016/j.probangmech.2007.12.026>.
- [83] G. Biau, D.M. Mason, High-dimensional p -norms, in: *Mathematical Statistics and Limit Theorems*, Springer International Publishing, 2015, pp. 21–40, http://dx.doi.org/10.1007/978-3-319-12442-1_3.
- [84] R.S. Anderssen, R.P. Brent, D.J. Daley, P.A.P. Moran, Concerning $\int_0^1 \cdots \int_0^1 (x_1^2 + \cdots + x_k^2)^{1/2} dx_1 \cdots dx_k$ and a taylor series method, *SIAM J. Appl. Math.* 30 (1) (1976) 22–30, <http://dx.doi.org/10.1137/0130003>.
- [85] V. Sadílek, M. Vořechovský, Evaluation of pairwise distances among points forming a regular orthogonal grid in a hypercube, *J. Civ. Eng. Manage.* 24 (5) (2018) 410–523, <http://dx.doi.org/10.3846/jcem.2018.5189>.
- [86] C.C. Aggarwal, A. Hinneburg, D.A. Keim, On the surprising behavior of distance metrics in high dimensional space, in: *Database Theory — ICDT 2001*, Springer Berlin Heidelberg, 2001, pp. 420–434, http://dx.doi.org/10.1007/3-540-44503-x_27.
- [87] A. Flexer, D. Schnitzer, Choosing ℓ^p norms in high-dimensional spaces based on hub analysis, *Neurocomputing* 169 (2015) 281–287, <http://dx.doi.org/10.1016/j.neucom.2014.11.084>.
- [88] M. Jamil, X. Yang, A literature survey of benchmark functions for global optimisation problems, *Int. J. Math. Model. Numer. Optim.* 4 (2013) 150–194, <http://dx.doi.org/10.1504/IJMMNO.2013.055204>.
- [89] M. Vořechovský, J. Mašek, Distance-based optimal sampling in a hypercube: Energy potentials for high-dimensional and low-saturation designs, *Adv. Eng. Softw.* 149 (2020) 102880, <http://dx.doi.org/10.1016/j.advengsoft.2020.102880>.
- [90] G. Migliorati, Adaptive approximation by optimal weighted least-squares methods, *SIAM J. Numer. Anal.* 57 (5) (2019) 2217–2245, <http://dx.doi.org/10.1137/18M1198387>.